# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| **1. Document** | | | | **P-78** | **P-78** | | | | | |
| 1 | Certification | 1 | Material certification to be provided. | Packager shall provide complete documentation according to Brazilian Labour and Employment Ministry NR-10. This documentation shall be issued and signed by legally qualified personnel. The documentation shall include the items listed below and all items foreseen in NR-10:<br>a) Conformity Technical Appraisal regarding applicable standards for all internal equipment and installations, including tests, inspections, and measurement reports;<br>b) Project Technical Report, including internal protection, installation, safety grounding, etc.;<br>c) Manuals with safety and health procedures and instructions regarding electrical equipment and installations;<br>d) List with electrical and electronic equipment installed in hazardous areas, including for each equipment, the classification Ex, nameplate data, number of Conformity Certificate and name of entity that issued the certificate;<br>e) Conformity Certificate for all electrical and electronic equipment proper for installation in hazardous areas, according to INMETRO Portaria nº 179, May 18th 2010 and Portaria nº 89, Feb 23rd 2012. | I-ET-3010.00-5140-700-P4X-003 | YES | | Noted | ONLY FOR APPLICABLE REQUIREMENTS TO OUR PACKAGE | Closed |
| | | 2 | Explosion proof certification to be provided. | The application of electrical equipment, instruments and accessories on hazardous areas shall follow the requirements of IEC series 60079, 61892-6 and 61892-7, as well as have their conformity certificates according to INMETRO Portaria 179, May 18th 2010 and INMETRO Portaria 89, Feb 23rd 2012 and approved by Classification Society.<br><br>Electrical equipment and accessories installed in external safe or hazardous areas, which shall be kept operating during emergency shutdown ESD-3P or ESD-3T shall be certified for installation in hazardous areas Zone 2 Group IIA temperature T3 (IEC 61892-7).<br><br>All hazardous areas equipment type certificates shall be delivered together with respective equipment.<br>All instruments that operate with electric power (transmitters, analyzers, switches, valve positioners and solenoids, fire detectors, gas detectors, sight gas detector beam emitters, manual call points - MCPs etc.) shall be certified to operate in zone 2.<br>The type of protection to be used is Ex dc IIA T3 in wide area and zone 2 and Ex db IIA T3 when installed in zone 1. The cases where another type of protection is acceptable are:<br>a) Manual call points, MCP;<br>b) Instruments installed in zone 0 (zero) shall have Ex ia IIA T3 certification.<br>Other types of protection with EPL Ga are not accepted.<br>c) For analyzers, where the only available technology is pressurized type, an air reservoir, adjacent and exclusive to the analyzer, shall be provided. It shall allow a minimum autonomy equal to the operating time of the CSS equipment defined in DR-ENGP-M-I-1.3 – SAFETY ENGINEERING If the protection type used is for zone 1, Ex pX, all circuits (DC power, AC power, discrete signal, analog signal etc.) shall pass through the disconnecting device. [INTERDISCIPLINARY INTERFACE] [PRO, MEC, HSE]<br>d) Protection type increased safety, Ex e, for junction boxes.<br>NOTE: Any other type of protection shall be submitted to PETROBRAS for evaluation. | I-ET-3010.00-5140-700-P4X-003<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Noted | IT IS NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA | Closed |
| | | 3 | Instrument calibration certification to be provided. | Instrument calibration certification to be provided.<br>1) Valid period shall meet the whole project schedule(Project handover) plus min 2 years.<br>2) Original of the calibration sheet shall be delivered before delivery. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 4 | WPS/PQR to be provided. | Weld certificates shall be provided attesting procedures, qualification of the performer and penetrating liquid testing of the weld in accordance with the technical specification document of the project entitled PIPING SPECIFICATION. This is general requirement for all instrument which needs welding. | I-ET-3010.00-1200-800-P4X-013 | YES | | Noted | NDE According to our standards | Closed |
| | | 5 | SIL certification to be provided<br>Reliability and Availability | SIL certification to be provided for Safety devices. | N/A (HHI Standard) | YES | | Noted | ONLY FOR A SAFETY RELAY IN THE MAIN CONTROL PANEL | Closed |
| 2 | Instrument input data | 1 | IO list and index<br>-IO list/ Index to be provided via project format. | All document and drawings shall be in PDF format and provided with its respective native files. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 2 | Mapping list / Serial IO list<br>-Mapping list to be provided via project format. | All document and drawings shall be in PDF format and provided with its respective native files. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 3 | Cable schedule<br>-Cable schedule to be provided via project format. | All document and drawings shall be in PDF format and provided with its respective native files. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 4 | Wiring/termination DWG<br>-Cable termination to be provided. | All document and drawings shall be in PDF format and provided with its respective native files. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 5 | Graphic layout<br>-Graphic layout to be provided if applicable. | To be provided as per requirement | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 6 | Cause and Effect chart /diagram<br>-ESD/FGS cause and effect chart to be provided as per project requirement | To be provided as per requirement. Specially, even though fire gas system is not a part of PKG system, FGS C&E shall be provided when it shall be protected by fire fighting system such as CO2 or watermist. | I-ET-3010.00-1200-800-P4X-012 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 7 | System architectures<br>-System architectures to be provided. | To be provided as per requirement | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 8 | Functional logic diagram<br>-Functional Logic Diagrams, Function Descriptions and Control Narratives to be provided. | To be provided as per requirement | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 9 | Instrument data sheet<br>-Instrument data sheet to be provided via project format. | Data sheet to be provided as per ISA format<br>All document and drawings shall be in PDF format and provided with its respective native files. | ISA | YES | | Noted | | Closed |
| | | 10 | Instrument loop DWG<br>-Instrument loop DWG to be provided.<br>-Instrument loop index shall be provided as per project requirement. | All document and drawings shall be in PDF format and provided with its respective native files. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 1 | Load list<br>-Power consumption calculation shall be provided as per project requirement. | Electrical Loads List<br>The following information shall be provided, as a minimum:<br>Project Template will be provided later.<br>- Load Identification (TAG);<br>- Load Identification (Description);<br>- Rated voltage (V);<br>- Rated power (kW for motor and kVA for other loads);<br>- Rated current (A);<br>- Starting time (s) and starting current (p.u.);<br>- Speed (rpm);<br>- Load required power (bkW);<br>- Motor plus load consumptions (kVA);<br>- Service factor;<br>- Efficiency (%) at 25, 50, 75 and 100% of load;<br>- Power factor at 25, 50, 75 and 100% of load;<br>- Maximum locked-rotor time (s). | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | | Closed |
| | | 2 | Heat dissipation calculation sheet<br>- Heat dissipation data shall be provided. | Heat dissipation calculation at maximum and normal operational conditions (W) shall be provided. | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | | Closed |
| | | 3 | Motor data sheet<br>-Requirement of maximum number of start of motor/ restart inhibit time at cold and hot condition shall be clearly indicated on motor datasheet by vendor, if required.<br>Motor data sheet including minimum below data in case MV motor shall be provided for power system calculation.<br>- Motor thermal over load curve (Hot/Cold) for Stator at 100% and 80% voltage<br>- Motor locked rotor withstand curve (Hot/Cold) for Stator at 100% and 80% voltage<br>- Motor Acceleration curve (current vs time) at 100% and 80% voltage<br>- Motor Acceleration thermal limit curve (Refer attached example as per IEEE C37.96-2012)<br>- Motor Load Torque curve at 100% and 80% voltage<br>- Motor and Load Inertia | As per I-LI-3010.00-5140-700-P4X-001 and I-ET-3010.00-5140-712-P4X-001(or I-ET-3010.00-5140-712-P4X-002) | I-LI-3010.00-5140-700-P4X-001<br>I-ET-3010.00-5140-712-P4X-001<br>(for LV Motor)<br>I-ET-3010.00-5140-712-P4X-002<br>(for MV Motor) | YES | | Noted | WE WILL PROVIDE THE INFORMATION AVAILABLE FROM THE SUB-SUPPLIER | Closed |
| | | 4 | THD harmonic spectrum data report<br>-THD value shall comply with project requirement<br>- To be provided. | The voltage distortions and the notch depth shall remain within the following limits, as stated in IEEE 519:<br>a) Total Harmonic Distortion (THD) – 5%;<br>b) Individual Harmonic Distortion – 3%;<br>c) Notch depth – 20%. | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 5 | Schematic diagram / Single line diagram<br>-To be provided for electric equipment | Schematic / Single line diagram diagram to be submitted for all electric equipment. | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| 3 | Elec. input data | 6 | Equipment layout<br>- Component, equipment location with tag<br>- Gland plate with dimension and cable<br>- Filed cable interface point for MCT and Tie-in cable tray.<br>- Earth connection point<br>- Recommended minimum clearance distance (side, front, rear and top) of equipment which is reqiured for proper operation, test and maintenance shall be indicated on general arrangement drawing.<br>- The door opening direction, Front & Rear direction and dimension for panels shall be indicated on general arrangement drawing.<br>- Weight data to be provided. | Equipment layout shall have the plans and its sections & details which depict, in the vertical plane, all sections and views indicated on the Equipment Layout Plan, including complete identification and dimensions of equipment.<br>- Component, equipment location with tag<br>- Gland plate with dimension and cable<br>- Filed cable interface point for MCT and Tie-in cable tray.<br>- Earth connection point<br>- Recommended minimum clearance distance (side, front, rear and top) of equipment which is reqiured for proper operation, test and maintenance shall be indicated on general arrangement drawing.<br>- The door opening direction, Front & Rear direction and dimension for panels shall be indicated on general arrangement drawing.<br>- Weight data to be provided. | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | | Closed |
| | | 7 | Connection diagram<br>-Connection DWG to be provided for interface cable | Packager shall present interface cables list and interconnection diagrams, clearly indicating the cables supplier (Packager, or Module Supplier, or Bidder). | I-ET-3010.00-5140-700-P4X-003 | YES | | Noted | | Closed |
| | | 8 | Electrical diagram and schematic dwg<br>-To include BOM with original manufacturer part number. | Electrical diagram and schematic dwg shall indicate BOM with original manufacturer part number. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 9 | Serial mapping address data for electric system interface<br>-It shall be submitted if required. | If any communication between eletric system and PKG, serial mapping list to be provided. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 10 | The detail list for all ship loose items shall be provided. The installation details like hook up and layout drawings for these items to be also included along with this list. | The detail list for all ship loose items shall be provided. The installation details like hook up and layout drawings for these items to be also included along with this list. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 11 | 3D modeling for entire package shall be done by the vendor and native files shall be provided to HHI. This 3D model shall include the layout for E&I items like panel, instruments, JB and trays within the package. All the clearances and dimensions shall be clear in the model. | 3D modeling for entire package shall be done by the vendor and native files shall be provided to HHI. This 3D model shall include the layout for E&I items like panel, instruments, JB and trays within the package. All the clearances and dimensions shall be clear in the model. | N/A (HHI Standard) | YES | | Noted | 3D MODEL OF THE PACKAGE IN STEP FORMAT | Closed |
| | | 12 | The MCT cable assignment<br>-MCT cable assigment to be provided including yard cables. | MCT cable assigment to be provided including yard cables. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 13 | Certificate of Conformity for Ex Electrical Apparatus<br>- To be provided. | 2.23. Certificate of Conformity for Ex Electrical Apparatus<br>2.23.1. The Certificate of Conformity related to a given Ex electrical apparatus shall be issued in a dedicated design document, coded in accordance with the requirements of NI-1710 and made available through a searchable PDF file.<br>2.23.2. A given Certificate of Conformity may be assigned to identical Ex electrical apparatus from the same subsystem/contractor. In this case, only one design document need to be issued. | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA | Closed |
| | | 14 | Ex Electrical Apparatus List | Packager shall furnish a table with the data presented in Table below for Ex Electrical Apparatus.<br>Equipment Tag. No. / Component identification<br>Equipment Description / Characteristics<br>Unit Location / Module<br>Equipment Manufacturer<br>Catalogue / Type Number<br>IP Code<br>Type of Ex Protection<br>Gas Group<br>EPL (Equipment Protection Level)<br>Temperature Class<br>Approval Body<br>Certificate Type<br>Certificate Number<br>Zone of Use<br>Accessories<br>Certificate of Conformity | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA | Closed |
| 7 | Reliability data for SIL | 1 | PKG vendor shall submit the reliability data for SIL verification via the project format. | To be provided. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 8 | Cyber security work sheet | 1 | Cyber security work sheet shall be submitted via project format if applicable. | *DETAILED ENGINEERING DESIGN DOCUMENTS specifed on I-MD-3010.1M-5520-800-P4X-003 _section 9 shall be supplied for review/approval.<br>9.2 NETWORK ONE-LINE DRAWING<br>9.3 DETAILED NETWORK INTERCONNECTION DIAGRAM<br>9.4 IP LIST<br>9.5 ASSET LIST<br>9.6 Other documents<br>A cybersecurity robustness testing shall also be performed. This test includes vulnerability analysis, penetration testing, and submitting the system to unusual network loads to ensure that it maintains its correct behavior. | I-MD-3010.1M-5520-800-P4X-003 | | NO | Not acceptable. eventhough no network connection with CSS, still remain other cybersecurity points such as hardening, password. | NOT APPLICABLE FOR OUR PACKAGE<br>14/07/2021: The bilge alarm in protected by a passwword, in order to avoid any changes in the configuration.<br>PLC is also protected by a password.<br>HHI20210716] Noted | Closed |
| 9 | E&I Tie in document | 1 | 1. E&I tie in document shall be provided as per E&I Tie document guidance.<br>2) E&I tie in document shall indicate the all information related to HHI yard cable interface.<br>3. The information related cable installed by vendor shall exclude from this document.<br>4. If any discrepancy between this doc and others, E&I tie in document shall be pripority for HHI yard interface finalization.<br>5. The contents of table for this document shall be<br>- Tie in Cable block diagram [ Project format ]<br>- Tin in Cable schedule [ Project format ]<br>- Tie in E&I layout<br>- Tie in panel layout<br>- Tie in installation detail for ship loose items | Refer to E&I tie in document guidance. | E&I tie in document guidance. | YES | | Noted | WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES | Closed |
| | | 2 | Tie in Cable block diagram<br>- It shall include all cables which are installead by HHI at yard.<br>- Cable installed by vendor within skid shall excluded in this document.<br>- The project format shall be used.<br>- All required data on format shall be filled out. | This document shall be submitted by employing project tie in document template / format. This template requests to fill out the following data and it is mandatory to fill out all required data as per project requirement/numbering procedure.<br>A. Cable tag numbering<br>B. Cable type<br>C. IO tag / type for instrument<br>D. Load description for electric<br>E. Power rating / consumption<br>F. From / To equipment tag number/description<br>G. Cable entry size | E&I tie in document guidance. | YES | | Noted | WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES | Closed |
| | | 3 | Tie in Cable schedule/ Tubing schedule<br>- It shall include all cable/tubing list which is installead by HHI at yard.<br>- Cable/tubing list installed by vendor within skid shall excluded in this document.<br>- Cable scheudle shall be generated based on tie in cable block diagram. | Tie in cable schedule shall be submitted after receiving 1st comment on cable block diagram from Bidder. It is Packager's responsibility not only to incorporate all comment /data on cable block diagram to this document but also to exclude cable list installed by VENDOR from this DWG.<br>In addition, it is required to submit it via tie in cable schedule format. | E&I tie in document guidance. | | | Noted | WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES | Closed |
| | | 4 | Tie in Cable wiring/termination<br>- It shall incldue all wiring/termination under HHI scope of work.<br>-Termination information under vendor scope shall exclude in this document. | Tie in termination DWG shall include following information as minimum.<br>A. Cable tag numbering<br>B. Cable type<br>C. IO tag / type for instrument<br>D. Power load / Power signal description for Electric<br>E. From / To equipment tag number<br>F. Terminal block number / Terminal block strip number<br>G. Yard cable screen/shield termination<br>H. Diode / resistance for line monitoring if applicable | E&I tie in document guidance. | | | Noted | WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES | Closed |
| | | 5 | Tie in E&I layout<br>-It shall indicate all cable tray/tubing tray that HHI yard cable/tubing to be laid.<br>-It shall indicate all equipment that HHI yard cable to be terminated.<br>-Cable benidng radius shall be visualized based on mximum yard cable OD be laid on each cable tray.<br>-List for yard cable laid on each cable tray shall be shown.<br>-Tray size information to be shown | This document is key data to determine yard cable routing. Following information shall be shown on package skid layout.<br>A. The location of E&I equipment.<br>B. Note: E&I equipment tag number shall be matched with the data on tie in cable block diagram<br>C. Cable tray with dimensional data between skid edge tie in location and E&I equipment.<br>D. Note: Only cable tray on which laid yard cable need to be shown.<br>E. Skid edge tie in location / points<br>F. Cable list with assumed in skid length / tubing list laid on each cable tray<br>G. Cable tray spare space data after installing yard cables. | E&I tie in document guidance. | | | Noted | WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES | Closed |
| | | 6 | Tie in panel layout<br>-Termination point for each yard cable to be shown in this DWG.<br>- Foot print / fixing hole size / panel dimension to be indicated.<br>- Door opening to be shown.<br>- Gland plate hole arrangement layout | It is critical document for the control room arrangement. It shall show the information of panel foot print with fixing holes detail, door opening, gland plate holes arrangement and termination points for each yard cables. | E&I tie in document guidance. | | | Noted | WE WILL PROVIDE ELECTRICAL DRAWINGS OF THE PACKAGE AND ACCORDING TO VDRL AGREED BY BOTH PARTIES | Closed |
| | | 7 | Tie in hook up / Installation detail<br>-It shall include the installation detail for all field instrument/detectors/electric devices that are supplied by vendor but are installed by HHI. | Tie in E&I installation detail shall indicate the mounting method for all loosed E&I equipment which is installed by CONTRACTOR(Bidder) at yard including required bill of material. .<br>It is crucial to indicate the mounting holes size on mounting bracket/plate as well as recommended mounting bolt sizes with torque value. | E&I tie in document guidance. | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| 10 | Project Document List | 1 | Documentation required in I-ET-3010.00-5140-700-P4X-003 clause 9 shall be submitted for approval. | Documentation required in I-ET-3010.00-5140-700-P4X-003 clause 9 shall be submitted for approval as a minimum. | I-ET-3010.00-5140-700-P4X-003 | | | | ACCORDING TO VDRL AGREED BY BOTH PARTIES | Closed |
| **2. General requirement** | | | | | | | | | | |
| 1 | Painting | 1 | All E&I devices shall be painted/coated as per project painting requirement. | 1. Electrical<br>Paint system for external coating of all electrical equipment, material and supports shall be proper for offshore installation and pre-qualified according to I-ET-3010.00-1200-956-P4X-002 - GENERAL PAINTING.<br>Painting system for internal coating of all electrical equipment and material shall be proper for offshore installation according to manufacturer requirements, when not specified by PETROBRAS.<br>Unless otherwise stated in equipment documentation, the last coat colour for electrical equipment shall be Light Green Munsell 5G8/4. Inner components mounting plate, internal doors faces shall be Safety Orange Munsell 2.5YR6/14.<br>Note: Colours shall comply with NR-37.<br>For carbon steel parts of lighting fixtures and junction boxes, the last coat colour shall be White Munsell N9.5.<br>For stainless steel lighting fixtures no coat colour is required.<br>For FRP materials, no coat colour is required, unless otherwise required by a standard, regulation or PETROBRAS.<br>For fire fighting associated equipment, the last coat colour shall be Safety Red Munsell 5R4/14.<br>Stainless steel surfaces shall be painted if exposed do saline atmosphere and temperatures above the limits indicated, as defined in I-ET-3010.00-1200-956-P4X-002 - GENERAL PAINTING:<br>● 316L - with service temperature over 50°C shall be coated;<br>● Duplex SS - service temperature over 80°C shall be coated;<br>● Superduplex SS - service temperature over 90°C shall be coated.<br><br>2. Instrument<br>Instrument shall follow the piping or equipment color.(DR-ENGP-I-1.15)<br>Instrument panel color shall be Light cream.<br>All instrument shall be painted as per I-ET-3010.00-1200-956-P4X-002. | 1. I-ET-3010.00-5140-700-P4X-002<br>2. I-ET-3010.00-1200-956-P4X-002<br>DR-ENGP-I-1.15 | YES | | Noted | | Closed |
| 2 | VCI Coating | 1 | VCI coating shall be applied as per project requirement. | Refer to AUTOMATION PANELS, I-ET-3010.00-5520-888-P4X-001 .<br>In order to prevent corrosion, VCI shall be used adequately, where applicable, as part of preparation for shipment and storage instead of desiccants such as silica gel. The latter shall be used only in cases where VCI is not applicable. Both VCI and desiccants shall not be used together for protecting the same compartment. | I-ET-3010.00-5520-888-P4X-001 | | NO | vendor to provide | NOT APPLICABLE FOR OUR PACKAGE<br>14/07/2021: noted, to be discussed in the meeting<br>HHI20210716] Noted | Closed |
| 3 | Cable glands | 1 | Cable gland type shall be selected as per project requirement. | Cable gland shall follow IEC 62444.<br>Cable glands for installation at non threaded holes of removable plates or steel sheet enclosures shall have cylindrical thread with locknut.<br><br>Unless otherwise stated, all threaded joints shall be taper type, NPT with standardized tolerances, according to ASME B1.20.1. Cylindrical threaded joints shall be acceptable for lighting fixtures for fluorescent lamps, for control boxes for push-buttons and signalling, for floodlights and for power socket-outlets. Cylindrical threaded joints for other applications shall be submitted to PETROBRAS approval. For equipment installed in hazardous areas, the threads shall comply with the requirements of IEC 60079-0.<br><br>Cable glands for armoured cables shall be metal type and the metallic pair shall not create an electrolytic corrosion in case of dissimilar metallic material. Stainless steel and copper armour shall use Stainless Steel Cable glands.<br>Barrier type Cable gland shall be supplied for Exd devices when IEC-60079 requests it. | I-ET-3010.00-5140-700-P4X-002 | YES | | Noted | | Closed |
| | | 2 | Material shall be selected as per project requirement. | Cable gland materials shall be selected in order to avoid electrolytic corrosion caused by contact of dissimilar materials.<br>Cable glands material shall follow the requirements of Table 18:<br><br>*Table 18 - Cable Gland Material*<br>Enclosure Material: Stainless Steel — Cable Gland Material: Stainless Steel AISI-316L<br>Enclosure Material: Cast Iron — Cable Gland Material: Aluminium<br>Enclosure Material: Aluminium — Cable Gland Material: Aluminium<br>Enclosure Material: FRP (Fiber Reinforced Plastic) — Cable Gland Material: Aluminium; Nylon (1); Naval Bronze (2); Stainless Steel AISI-316L<br>Enclosure Material: Stainless Steel, Naval Bronze — Cable Gland Material: Naval Bronze (2)<br>Notes: 1) Nylon cable glands shall be accepted only up to maximum size 1", with metallic plate (made of Stainless Steel AISI-316L, Aluminium or Naval Bronze(2)) for grounding, with internal locknut and if they are certified as Ex e or Ex n.<br>2) Naval Bronze shall be classic marine, high-strength and corrosion-resistant alloy C-462 or C-464. | I-ET-3010.00-5140-700-P4X-002 | YES | | Noted | | Closed |
| | | 3 | Ex certification shall be provided as per project requirement | Conformity Certificate for all electrical and electronic equipment proper for installation in hazardous areas, according to INMETRO Portaria nº 179, May 18th 2010 and Portaria nº 89, Feb 23rd 2012. | I-ET-3010.00-5140-700-P4X-002 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA | Closed |
| | | 4 | IP washers shall be installed as per cable gland manufacturers' recommendation and project requirement.Fabric or rubber type IP washer is prohibited. | To be provided. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 5 | Serrated washers shall be provided as per project requirement. | To be provided. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 6 | Cable gland for HHI yard cable interface shall be provided by PKG vendor.<br>Cable detail information to be provided by HHI during detailed engineering. | To be provided. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 7 | All spare cable entries shall be plugged with Ex certified cable plug. | To be provided. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE, SINCE IT IS LOCATED IN SAFE AREA | Closed |
| | | 8 | Cable gland shroud shall be provided as per project requirement. Long type shroud to be selected in case cable gland fitting (reducer, adoptor) is applied to cover all part of gland and accessory | N/A | N/A | | | Noted | N/A | Closed |
| | | 9 | Cable Gland Code List | Enforce that CONTRACTOR's sub-suppliers and PACKAGED UNIT's SUPPLIERS also create such codes and emit a similar document. | I-ET-3010.00-1200-800-P4X-010<br>Refer to Form-10 | | | Bidder to check | 14/07/2021: NOTED | Closed |
| 4 | INST. Hook up | 1 | Hook-up drawing to be provided by PKG vendor. | To be provided by VENDOR | API RP 551 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 2 | Tubing/tube fitting material, size, type, selected vendor to be in line with project requirement. | Tubing shall be made of ASTM A 269 Gr. TP 316L stainless steel with a minimum Molybdenum content of 2.5%.<br>The material of the associated connections and valves shall be made of AISI 316 stainless steel, not requiring a minimum Molybdenum content of 2.5%.<br>Note: In cases where the atmosphere (microclimate) is unsuitable for tubing in the material described in item I-ET-3010.00-1200-800-P4X-013 _ 19.2.1, the use of tubing, fittings and associated valves in the 6Mo stainless steel material (ASTM A269 S31254) may be considered.<br>Each batch of tubing shall have a Positive Material Identification certificate in accordance with API RP 578.<br>Connections shall have certificate of conformity with ASTM F1387. The certificates shall comprise at least the Standard Qualification Tests specified on 19.1.6.2 _I-ET-3010.00-1200-800-P4X-013.<br>(A) and the following Supplementary Tests (S):<br>The wall diameters and thicknesses which may be used are presented in Table 19.2-I_I-ET-3010.00-1200-800-P4X-013<br>Tube fitting shall be supplied by manufacturer who will be determined by CONTRACTOR during Detailed Engineering.<br>Tentative approved supplier is SWAGELOK, PARKER, HY-LOK. So, the highset cost to be considered for bidding. | I-ET-3010.00-1200-800-P4X-013 | YES | | Noted | ACCORDING TO OUR STANDARDS | Closed |
| | | 3 | All entry size and type for tubing connection shall be clearly specified on hook-up DWG. | All entry size and type for tubing connection shall be clearly specified on hook-up DWG. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 4 | If yard tubing to be connected to PKG skid or devices by HHI, it shall be included on hook-up DWG. | If yard tubing to be connected to PKG skid or devices by HHI, it shall be included on hook-up DWG. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 5 | Instrument manifold and isolation valve shall be provided as per project requirement. | Instrument manifold and isolation valve shall be provided as per project requirement. | I-ET-3010.00-1200-800-P4X-013 | YES | | Noted | | Closed |
| | | 6 | Vent/drain line with cap for instrument venting shall be provided as per project requirement | Vent/drain line with cap for instrument venting shall be provided as per project requirement | N/A (HHI Standard) | | | | N/A | Closed |
| | | 7 | The slope for tubing line shall be inline with project requirement as well as API PR 551. | When measuring liquids, impulse lines shall have a downward slope of 1:10 from the impulse tap up to the manifold block of the instrument. When measuring gas, the impulse lines shall have a rising slope of 1:10 from the impulse tap up to the manifold block of the instrument. | I-ET-3010.00-1200-800-P4X-013 | YES | | Noted | ACCORDING TO OUR STANDARD DESIGN FOR PROPER OPERATION | Closed |
| | | 8 | Open end for tubing line shall be provided with cap, bug screen, flame arrestor and etc as per project requirement. | Open end for tubing line shall be provided with cap(calibration connection/drain), bug screen(safety gas venting), flame arrestor(flamable gas venting). | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |

# Technical Compliance Sheet

| | |
|---|---|
| PROJECT : | P-78 |
| RFQ No.: | HQ-54 |
| EQUIPMENT NAME : | OILY WATER SEPARATOR |
| INSPECTED ON : | |
| INSPECTED BY : | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| | | 9 | Tubing clamp/ tray shall be provided as per project requirement. | When routing more than one tubing in parallel, appropriate insulation shall be provided to avoid contact between them. To avoid corrosion points, this insulation cannot create regions that can accumulate water<br>Support systems that can create crevice corrosion points shall not be used, such as strip-type or clamp-type supports. Tubing support systems shall be designed to minimize the points of contact between the tubing and the support to avoid crevice corrosion points.<br>The metal-to-metal contact shall be eliminated using insulating material for the parts in contact with the tubing.<br>Parallel tubing routing shall be done in covered trays and provision shall be made to avoid water accumulation. | I-ET-3010.00-1200-800-P4X-013 | YES | | Noted | | Closed |
| | | 10 | Sealant material shall meet project requirement. | All thread connection shall be sealed with sealant material. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 11 | Leak test report or mechanical completion check sheet for instrument hook up shall be provided. | Leak test report and mechanical completion check sheet shall be provided. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 12 | Tubing Support | The length of unsupported tubing to final destination (such as control valve or transmitter) for single tubes shall not exceed 0.5m. | N/A (HHI Standard) | YES | | Noted | | Closed |
| 5 | Labeling on Elec. Equipment | 1 | All E&I devices such as eletrical panels, juction boxs, field instruments and etc shall be provided with rigid tag plate as per project requirement | Identification nameplates for equipment with specific Technical Specification (e.g. induction motors, generators, MCCs and switchgears) shall follow the requirements of that specification.<br>All other equipment shall have 02 (two) identification nameplates.<br>The first plate shall have the equipment tag number, in black acrylic engraved with white letter for equipment installed indoors and in stainless steel AISI-316 for equipment installed outdoors.<br>The other plate shall be in stainless steel AISI-316 with the following information, when applicable, in Portuguese language:<br>Petróleo Brasileiro S.A. – PETROBRAS;<br>Fornecedor (vendor name);<br>Número de Série (equipment serial number);<br>Ano de fabricação (year of manufacture);<br>Número do pedido de compra (purchase order number);<br>Other equipment specific data;<br>Fabricante (manufacturer name);<br>Tipo (type);<br>Padrão de desempenho (performance standard);<br>Valores nominais para: potência aparente de saída, tensões, frequência, correntes (rated values for: output apparent power, voltage(s), frequency, currents);<br>Classificação térmica da isolação (thermal classification of insulation);<br>Grau de proteção IP do invólucro e caixa de terminais (IP code of enclosure and termination box);<br>Grupo de ligação de bobinas (vector group of windings);<br>Máxima temperatura admissível para o meio refrigerante (maximum permissible cooling medium temperature);<br>Valores das impedâncias de curto-circuito (short-circuit impedances values);<br>Tipo do meio refrigerante (type of cooling medium);<br>Massa total (total mass).<br>Cable trays and channels shall have body identification.<br>AISI 316 stainless steel identification plate with at least, the information specified on I-ET-3010.00-1200-800-P4X-013.<br>Besides the identification plate described in item 6.1.1, a stainless steel plate with a minimum size of 50 mm x 20 mm x 0.5 mm (Width x Height x Thickness) shall be provided with instrument respective TAG engraved as per I-ET-3010.00-1200-800-P4X-013. | I-ET-3010.00-5140-700-P4X-002<br>I-ET-3010.00-1200-800-P4X-013 | YES | | Noted | | Closed |
| | | 2 | All E&I devices shall be provided with location tag plate as per project requirement | For instrument at field [ To be considered for bidding ]<br>UV resistant weather proof laminated engraved traffolyte location tag plate to be supplied on instrument support so that when the instrument is replaced, the nameplates shall remain on the support. | N/A (HHI Standard) | | NO | Bidder to comply on this requirement except UV resistant weather proof requirement. | NOT APPLICABLE FOR OUR PACKAGE<br>14/07/2021: there are not field instruments in our package. However, we will follow your requirement for the instruments within our package<br>HHI 20210716] Noted | Closed |
| | | 3 | Tag plates for electrical components such as power supply, IO card, terminal strips, relay, Earthing bus bar and etc inside panel / junction box shall be provided as per project requirement.<br>Fuses shall be labeled with their current ratings. If the color coded wiring to fuses connected to the main circuit is not visible from the front of the fuses, the fuses shall also be labeled with their phase color. | Laminated engraved traffolyte name plate to be provided in order to identify each component.<br>Nameplates shall be provided for all devices, accessories, equipment, switches, etc., installed on the front plate or inside the panels, adjacent to each component, so that when the components is replaced, the nameplates shall remain in the panel.<br>Identification shall be in Brazilian Portuguese.<br>The grounding bars shall be clearly and permanently identified with nameplates in Brazilian Portuguese. The nameplates shall be made of black acrylic material and white lettering.<br>The identifications to be engraved shall be "ATERRAMENTO DE SINAL" for instrument grounding, "ATERRAMENTO DE SEGURANÇA" for safety grounding and "ATERRAMENTO DE SEGURANÇA INTRÍNSECA" for intrinsic safety grounding.<br>Identification of panels and internal equipment shall follow Brazilian standard regulation NR-12 - SEGURANÇA NO TRABALHO EM MÁQUINAS E EQUIPAMENTOS. | I-ET-3010.00-5520-888-P4X-001 | YES | | Noted | | Closed |
| | | 4 | Tubing tag plate shall be provided as per project requirement | Tubing tag plate shall be provided. | I-ET-3000.00-1200-940-P4X-001 | YES | | Noted | | Closed |
| | | 5 | Wire marker shall be provided as per wire marker numbering procedure. | Wiring shall be identified by heat-shrinkable ferrules or plastic rings, at each end, with numbers allocated by the Manufacturer. All wiring drawings shall clearly identify the appropriate reference numbers. For bidding, highest cost between heat-shrinkable ferrules and plastic rings shall be considered. During detailed engineering, type of wire marker shall be approved by CONTRACTOR.<br>All panel wiring shall be identified at both ends by plastic rings with the terminal block number.<br>Wire numbering procedure to be confirmed during detailed engineering. | I-ET-3010.00-1200-800-P4X-002 | YES | | Noted | | Closed |
| | | 6 | Multi cable transit(MCT) shall be provided with tag plate. | 1) Tag Plate : Each side of frame<br>2) Material : AISI-316L stainless steel<br>3) Tagging procedure as per clause 3.10.4 of I-ET-3010.00-5140-700-P4X-001 | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | IF APPLICABLE | Closed |
| | | 7 | Cable tag plate shall be provided as per project requirement | All cables shall be clearly identified by means of plastic rings for identification or other method that supports the operation conditions, with the design number, at least at the following points:<br>a) at panels, boxes and equipment entrance;<br>b) at both sides of MCTs and any bulkhead crossing point incomings and outgoings;<br>c) at the ends of conduits feeding single equipment. | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | | Closed |
| | | 8 | All inscriptions on nameplates shall be in the English language and Language required in Project. | Both English language and Brazilian Portuguese language to be provided for applicable items as per project specification. | I-ET-3010.00-5140-700-P4X-002<br>I-ET-3010.00-1200-800-P4X-013 | YES | | Noted | | Closed |
| | | 9 | Danger and warning labels shall comply with IEC 60417 and ISO 3864. | All electrical equipment, flour mounted, panels, or similar in construction to a panel, regardless of the area where it is installed, shall have the warnings as required by NR-10 and NR-12.<br>Warnings shall follow the standard labels as required in ABNT NBR 13434-2 for electrical panels risk of shock as informed in I-ET-3010.00-5400-947-P4X-002 - SAFETY SIGNALLING.<br>Power Panels shall have warning labels indicating the protective clothing risk category that shall be used for technical intervention.<br>Power Panels shall have warning labels indicating that any technical intervention in the panels shall be executed only by authorized people.<br>All Ex equipment which need to be operated during ESD condition shall be marked for easy identification with "ESD" label, including Ex equipment installed in safe area (outside hazardous area). | I-ET-3010.00-5140-700-P4X-002 | YES | | Noted | | Closed |
| | | 10 | Nameplates shall be permanently fixed with the screws, rivets, or nuts and bolts. Paper labels fixed with adhesives are not acceptable. | Nameplates shall be permanently fixed with the screws, rivets, or nuts and bolts. Paper labels fixed with adhesives are not acceptable. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 11 | All items of equipment shall have suitably inscribed identification labels indicating tag number and function. In addition, control stations and similar equipment shall carry normal operating function labels i.e. "stop", "start", "on", "off", etc. | All items of equipment shall have suitably inscribed identification labels indicating tag number and function.<br>For more details, refer to I-ET-3010.00-5140-700-P4X-005 REQUIREMENTS FOR HUMAN ENGINEERING DESIGN FOR ELECTRICAL SYSTEMS OF OFFSHORE UNITS. | I-ET-3010.00-5140-700-P4X-005 | YES | | Noted | | Closed |
| | | 12 | Withdrawable parts shall be equipped with an engraved identification nameplate. | Withdrawable parts shall be equipped with an engraved identification nameplate. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 13 | Labels shall not be fixed to components which may be replaced or removed, unless a second label is fitted internally to ensure correct replacement. | Labels shall not be fixed to components which may be replaced or removed, unless a second label is fitted internally to ensure correct replacement. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 14 | Certified enclosures are not to be drilled or altered in any way for the addition of labels. If labels are found to be missing on certified equipment, then the Supplier shall refer to their sub-Supplier or manufacturer for corrective action. | Certified enclosures are not to be drilled or altered in any way for the addition of labels.<br>If labels are found to be missing on certified equipment, then the Supplier shall refer to their sub-Supplier or manufacturer for corrective action. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 1 | Earthing bar, boss, plate, stud shall be provided for yard earthing cable.<br>Ex) Junction box, vessel, skid, motor, panel, field devices…… | Requirement in section 3.13 of I-ET-3010.00-5140-700-P4X-001 SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS shall be fulfilled.<br><br>AISI 316L Grounding bar and grounding copper cable inside package shall be provided by Packager.<br>For details, refer to I-DE-3010.00-5140-700-P4X-003 GROUNDING INSTALLATION TYPICAL DETAILS. | I-ET-3010.00-5140-700-P4X-001<br>I-DE-3010.00-5140-700-P4X-003 | YES | | Noted | | Closed |
| | | 2 | Equipment door (JB, cabinet and etc) hinges shall be heavy duty type to support the door when opened. Earthing shall be accomplished by means of an external/internal earth stud, which can be connected to the terminal mounting rail or compound mounting plate. | Vertical sections shall have hinged front door. Doors shall be bonded to the panel structure through flexible copper cable. Hinged doors shall have an open position lock device. | I-ET-3010.00-5140-700-P4X-002 | YES | | Noted | IF APPLICABLE | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| 6 | Earthing | 3 | All non-current carrying metallic parts which may develop to electrical potential from any electrical component shall be earthed by a separate insulated earth conductor of suitable cross section. | All non-current carrying metallic parts which may develop to electrical potential from any electrical component shall be earthed by a separate insulated earth conductor of suitable cross section. | I-ET-3010.00-5140-700-P4X-002 | YES | | Noted | | Closed |
| | | 4 | The following items shall be earthed to the skid frame:<br>- Motor frames and generator frames<br>- Metallic armoring, cable troughs, wireways and tray, equipment racks, and busways.<br>- The armor or cladding on all cables shall be earthed at both ends. | Metallic structures, equipment, cables and accessories shall be grounded in accordance with the requirements of IEC 61892-6 and Classification Society. For FPSO and FSO units, the requirements of IEC 60092-201 and IEC 60092-502 shall also be complied with for grounding of equipment, cables and accessories.<br>In hazardous areas, the requirements of IEC 61892-7 shall be additionally complied with.<br>The grounding installation shall equalize all extraneous conductive metallic parts (conductive part not forming a part of the electrical installation and liable to propagate a potential, including ground potential) at the same potential to ensure that danger to operator is minimized.<br>All grounding connections shall be made with self-locking screws (connectors). Welded terminals or exothermic connections are permitted only between earth bosses and structure/skid. | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | | Closed |
| | | 5 | Metalic pipe flange shall be earthed to ensure electrical continuity to avoid any static current.<br>Pipe earhthing detail shall be included in offical vendor data for approval. | Bonding jumpers shall be installed across metal piping flange connections provided with non-conductive gaskets between them, regardlessof the kind of fluid that flows inside the piping.However, if RTJ (Ring Type Joint) metal gaskets or equivalent are used, the interconnection of flangeswith bonding becomes unnecessary.In either case (need of bonding jumpers or use of RTJ metal gaskets), the whole structure shall be in compliance with the requirements below.<br><br>Unless otherwise stated in this specification, metallic structures (equipment, vessels, tanks, piping, etc.) for use in fluid handling applications,when directly welded or bolted to the main plataform structure, need no additionalconnections to earth,provided the resistance to ground, as measured with a DC multimeter at any point of the said metallic structures, does not exceed 10 Ω. Measurements shall be carried out at the final stage of the delivery, aiming at keeping the resistance to ground below 106 Ω over the lifetime of the offshore unit, according to IEC/TS 60079-32-1. | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | | Closed |
| | | 6 | Earth bar inside panel and JB shall accommodate all earth connetion with interface cable fully. This earth bar hole q'ty and arrangement dwg shall be included in GA dwg. | Earth bar inside panel and JB shall accommodate all earth connetion with interface cable fully. This earth bar hole q'ty and arrangement dwg shall be included in GA dwg. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 7 | Earth bar type shall be bus bar type with drilled hole. Special type bus bar is not acceptable. | Earth bar type shall be Stainless steel AISI 316L flat bus bar type with drilled hole and installed directly using weld. Special type bus bar is not acceptable. | N/A (HHI Standard) | YES | | Noted | ACCORDING TO OUR STANDARDS | Closed |
| | | 8 | Cable tray bonding shall be provided as per project requirement. | Cable tray bonding shall be provided as per project requirement, refer to I-DE-3010.00-5140-700-P4X-003_E_GROUNDING INSTALLATION TYPICAL DETAILS. | I-DE-3010.00-5140-700-P4X-003 | YES | | Noted | ACCORDING TO OUR STANDARDS | Closed |
| | | 9 | Earthing for unit control panel /Elec. cabinet<br>1. Earthing shall be provided as per project requirement.<br>2) Proper earthing bus bar (Qty, bolt size....) shall be provided for yard earthing cables.<br>3. Earthing bus bar material/size shall be in line with project requirement<br>4. Maximum 2 earth wire connections are allowd per one earth bus bar bolt.<br>5. Dedicated earthing bus bar for each earthing system such as IE, ISE and SE shall be provided as per project requirement.<br>6. Earthing method for instrument cable shield shall be provided as per project requirement. | 1. All panels shall be supplied with internal grounding busbars and all panel metallic parts, which are not intended for current conduction, shall be interconnected to the grounding bar, including movable parts.<br>For the details, refer to I-ET-3010.00-5140-700-P4X-002_SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS clause 4.8.3.6 & 4.8.3.21.<br>2. Proper earthing bus bar (Qty, bolt size....) shall be provided for yard earthing cables which to be confirmed by vendor drawing.<br>3. Size of earthing bus bar shall be designed to accomodate total number of earthing cables to be connected.<br>4. Only one earth wire connection is allowd per one earth bus bar bolt.<br>5. Dedicated earthing bus bar for each earthing system such as IE, ISE and SE shall be provided as per project requirement.<br>6. Earthing method for instrument cable shield shall be provided as per project requirement | I-ET-3010.00-5140-700-P4X-002 | YES | | Noted | ACCORDING TO OUR STANDARDS | Closed |
| | | 10 | VENDOR shall provide Earthing boss/plate which is suitable for M10.<br>Otherwise, required size shall be specifed on relevant document. | VENDOR shall provide Earthing boss/plate which is suitable for M10.<br>Otherwise, required size shall be specifed on relevant document. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 11 | Tubing Earthing | Hydraulic pipes and tubings shall hasve electrical continuity and shall be grounded according to classification society rules. | Claasification Society | YES | | Noted | | Closed |
| 7 | Cable tray | 1 | Tray segregation shall be provided as per project requirement | Refer to P-78 CABLE TRAY SEGREGATION TABLE in folder 7. Other project key requirement. | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | | Closed |
| | | 2 | Tray tie in location for yard cables to be provided with PDMS model and tray layout.<br>Special attention for yard cable bending radius to be provided for the design of tie in point . | Tray tie in location for yard cables to be provided with PDMS model and tray layout.<br>Special attention for yard cable bending radius to be provided for the design of tie in point | N/A (HHI Standard) | | | Noted | 3D MODEL OF THE PACKAGE TO BE PROVIDED IN STEP FORMAT | Closed |
| | | 3 | Cable bending radius shall be reviewed at each cable entry, tie in points and bending routing. . | Cable bending radius shall be reviewed at each cable entry, tie in points and bending routing. . | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 4 | Cable tray for interface cable shall be supplied / installed at the skid edge by PKG vendor.<br>All internal cable route in package between skid edge and package equipment shall be prepared by vendor. | Cable tray for interface cable shall be supplied / installed at the skid edge by PKG vendor.<br>All internal cable route in package between skid edge and package equipment shall be prepared by vendor and identified in vendor drawing. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 5 | Cable tray type, material, painting shall be inline with project requirement. | 1) Cable trays for outdoor use shall be made from heavy duty stainless steel AISI 316L.<br>2) Cable trays for indoor use shall be either stainless steel AISI-316L or carbon steel painted according to clause 3.6.3 of I-ET-3010.00-5140-700-P4X-002.<br>3) All screws, nuts and washers shall be made of bichromatized steel or AISI-316L stainless steel. For application in cable trays and cable trays supports only AISI-316L stainless steel shall be used.<br>4) Requirement for cable tray specified in clause 5.3 of I-ET-3010.00-5140-700-P4X-002 to be fulfilled. | I-ET-3010.00-5140-700-P4X-002 | YES | | Bidder to comply with I-ET-3010.00-5140-700-P4X-002. | ACCORDING TO OUR STANDARDS<br>14/07/2021: Cable trays and bolting to be made SS316L | Closed |
| | | 6 | Minimum of spare capacity shall be provided in cable tray as per project requirement. | Cable trays, channels and ladders shall be designed with a reserve section of at least 20% of the total cross-section. | I-ET-3010.00-5140-700-P4X-001 | YES | | Noted | | Closed |
| | | 7 | Cable tray overfill to be verified by VENDOR. Cable tray overfilll shall be calculated considering future space requirement as well as yard cables.<br>Yard cables are cables which is installed at HHI yard/site by CONTRACTOR . | Trays, channels and ladders sizing calculation report and list shall be submitted for approval, refer to the details in clause 2.13 of I-ET-3010.00-5140-700-P4X-001 SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS. | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 8 | Cable tray routing shall be designed to maintain the mechanical continuity. If any deviation, it shall be approved via official documentation.<br>Supports for cable tray shall be provided to ensure to withstand the maximum cable load including future load. | Cable tray routing shall be designed to maintain the mechanical & electrical continuity. If any deviation, it shall be approved via official documentation.<br>Supports for cable tray shall be provided to ensure to withstand the maximum cable load including future load. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 9 | Cable tray cover shall be provided as per project requirement. And the cover shall withstand wind speed required in project operating environment condition. | 1. Cables within trays that transit decks and floors shall be protected from mechanical damage by installation of tray covers between grade level and up to 2m above grade.<br>2. Cable tray cover same material as the tray shall be provided, where required. i.e from exposure to sun, heat and mechanical damage.<br>3. Cable trays for voltage 6.6kV and above shall be fully covered reagradless areas. | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | NOT APPLICABLE, PACKAGE LOCATED INDOORS | Closed |
| | | 10 | If tubing line is installed between equipment within skid and skid outside, tubing within skid should be completed by vendor and prepared interface point edge of skid. Interface size and connection detail to be specified on hook up DWG. | If tubing line is installed between equipment within skid and skid outside, tubing within skid should be completed by vendor and prepared interface point edge of skid. Interface size and connection detail to be specified on hook up DWG. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 11 | Other cable tray requirement. | Cable tray accessories must be such as to permit the execution of the minimum radius of curvature of the biggest cable installed thereon.<br>Spacing and characteristics of the tray supports shall be determined so that they shall not be subjected to permanent distortion (allowing for 120% of installed cables), and also taking into account an 80kg individual point supplementary load applied at any point. Supports holding up the cable trays and conduits shall have a maximum spacing of 3 meters between them.<br>A minimum clearance of 300mm shall be kept between cable trays and heated piping or engine exhaust lines in case of parallel runs and 150mm in case of a cross-over. | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| 8 | Junction box | 1 | Material, painting and type for junction box shall be in line with project requirement. | Junction boxes enclosures shall be made of stainless steel AISI -316L and "Ex e" or "Ex de".<br>Junction boxes shall be provided with terminal blocks suitable for interconnection.<br>All metallic junction boxes shall comply with painting criteria of clause 3.6.3 of I-ET-3010.00-5140-700-P4X-002. | I-ET-3010.00-5140-700-P4X-002<br>I-ET-3010.00-1200-800-P4X-002 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 2 | It shall comply ot IP/Explosion poof requirement. | All outdoor JBs enclosures shall be IP-56 minimum ingress protection degree, according to IEC 60529. | I-ET-3010.1M-1200-800-P4X-005 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 3 | Jumper bar for cable shield shall be provided as per project requirement.<br>It shall include jumper bar for yard multi cable interface. | Jumper bar to be provided for shield termination of yard multi cables. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 4 | Dedicated junction box for each system/signal ( Ex. Analogue, digital, electric..) shall be provided as per project requirement. | Signals to CSS related to PSD/HSD and FGS/HFGS sub-systems shall be segregated in different terminal strips and multicables (where applicable), when installed inside the same junction box.<br>Cables from field instruments shall converge to junction boxes, segregated into 3 (three) types:<br>i. for signals using fire-resistant cables;<br>ii. for signals related to safety interlocks;<br>iii. for signals related to process control.<br>Junction boxes interconnecting fire-resistant cables and multicables shall have passive protection to provide the same fire resistance as the cables. The terminal strips shall also be of a fire-resistant material, such as ceramic.<br>The junction box shall be internally divided into two regions, one for discrete signals and one for analog signals.<br>Intrinsic safety signals shall be conditioned in a separate region inside the junction box, segregated from the other regions by a separating septum and identified as intrinsically safe signals. The terminal strips shall be blue.<br>"Ex e" or "Ex d" certification and certified to work according to the area classification where it is installed. In open areas, the minimum ertification shall be for zone 2 (EPL Gc). | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 5 | Top cable entry shall be probibitted. Bottom cable entry for multi cable is preferred. Side cable entry depends on project requirement. | Cables or multicables connecting instruments to the junction box may only be attached to the side or bottom of the junction box.<br>Multicables that connect the junction box to a panel or to another junction box shall be connected to the bottom of the junction box.<br>Entry of cables through the top of the junction box is not allowed. | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 6 | Cable entry hole/gland shall be provided according to yard cable information. | Cable entry hole/gland shall be supplied by vendor according to yard cable information to be provided during detailed engineering.<br>JBs shall be ample sized, with minimum of 20% spare (Terminals, cable entries). | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 7 | Removable gland plate shall be provided as per project requirement. | All enclosures shall have removable cable gland plates, If MCT is not required on project requirement.<br>For instrument JB,<br>In the lower part of the junction boxes, the passage of cables and multicables shall be done using a removable blind plate. No removable blind plate shall be used at the side entrance. | I-ET-3010.1M-1200-800-P4X-005 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 8 | Minimum spare capacity of terminal block shall be provided as per project requirement.<br>Terminal blocks shall be provided to terminate all yard multicores cables. | The terminal strips used inside the junction boxes shall be firmly attached and suitable for use in vibrating conditions to avoid poor contact and consequent spark. | I-ET-3010.00-1200-800-P4X-013 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 9 | Door hinge, cable duct shall be provided as per project requirement if applicable. | Door hinge, cable duct shall be provided as per project requirement if applicable. | N/A (HHI Standard) | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 10 | Spare cable entries shall be provided as per project requirement. All spare entries shall be plugged with Ex certified cable plug. | JBs shall be ample sized, with minimum of 20% spare (Terminals, cable entries)<br>MCTs shall be supplied with a spare of 10%.<br>All spare entry shall be pluged with certified plug. | I-ET-3010.1M-1200-800-P4X-005 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 11 | In case of power Junction box, voltage warning labels shall be installed on door cover. | Refer to clause 3.7 of I-ET-3010.00-5140-700-P4X-002 | I-ET-3010.00-5140-700-P4X-002 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 12 | Space for yard cable installation with bending radius should be reserved. | Space for yard cable installation with bending radius should be reserved. | N/A (HHI Standard) | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 13 | Cable entry thread type shall be in line with project requirement. | For all other cases, the threaded joints shall be taper type, NPT with standardized tolerances, according to ASME B 1.20.1. | I-ET-3010.1M-1200-800-P4X-005 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 14 | Earth continuity plate shall be provided in cable entries when non-metallic junction boxes are used. | Terminals strips, connectors and bolts used inside the JBs, located in Zone 2 shall be adequate for use when vibrations are present and avoiding sparks due to bad contact.<br>Instrument<br>Where required, the JBs shall have 1 (one) ground bar for grounding the armoring of cables. This bar shall be internally wired to the ground terminal of the box and it shall be provided with sufficient screws for terminating armor grounding wires. Each gland or gland plate shall be electrically bonded to its relevant equipment ground bar or terminal /junction box grounding stud.<br>For discrete instrument's cables, shield drain wires from instruments shall be interconnected in the junction box, nevertheless shall only be grounded on a dedicated ground bar for shield drain wires on the control panel.<br>For Analogic instrument's cable shield drain wires shall be connected to its respective multicable pair/triad/quad individual shield drain wire and then connected to a dedicated grounding bar shield drain wires on the control panel. Shield drain shall be electrically continuous from the instruments until the control panel.<br>he general shield drain wire on multicables shall be connected to dedicated grounding bar shield drain wires on the control panel and shall be left unconnected at junction box according to IEC-60079-14.<br>Each outdoor junction box shall have a ground bolt at the outside for bonding to the skid structure. This bolt shall terminate at the inside to provide a grounding means inside the junction box to the safety ground bar. | I-ET-3010.1M-1200-800-P4X-005 | | NO | Noted | NO JUNCTION BOXES IN OUR PACKAGE | Closed |
| | | 1 | General<br>1) Scale unit shall be employed with project requirement.<br>2) Mounting accessaries shall be provided as per project requirement.<br>3) If any dissimilar metal contact during installation is expected, insulation to be provided to avoid galvanized corrosion.<br>4) Accessibility shall be provided as per project requirement. | The following units for the main variables shall be used:<br>• Temperature: C<br>• Flow rate Liquids: m³/h<br>• Flow rate Water Vapor t/h<br>• Flow rate Gas: m³/h (@ 20 °C and 101.325 kPa abs)<br>• Pressure: bar-g or kPa-g<br>• Vacuum and Low Pressures: bar-a or kPa-a<br>• Level: % of the range, m or mm<br>Field insturment shall be supplied with mounting bracket and insulation shall be provided to void galvanized corrosion. | I-ET-3010.00-1200-800-P4X-002 | | NO | Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser. | NO FILED INSTRUMENTS IN OUR PACKAGE<br>21/07/2021: noted<br>HHI20210716] Noted. However, Vendor to follow MR requirement for the instruments within vendor package | Closed |
| | | 2 | Pressure transmitters<br>1) Housing<br>2) SIL<br>3) Material (wetted part Vs non wetted part)<br>4) Cablibration range<br>5) Accuracy / Repeatability / Temperature Effect<br>6) Instrument manifold if applicable<br>7) IP/ Ex. Proof.<br>8) Local display<br>9) Protocol ( Hart, FF, PB..)<br>10) Asset management interface configulation<br>11) Capillary tube protection if applicable<br>12) IS/Non IS<br>13) Flush ring with nipple & valve if applicable.<br>14) Cable Entry Size | 1) ASTM A351 Gr. CF8M stainless steel<br>2) SIL 2 for Safety (Bidding)<br>3) Min SS316L for wettted part, In cases where the process is unsuitable for SS316L, higher grade material.<br>4) Min 10% higher than high alarm/ trip set point / Min 10% lower than low alarm / Trip set point<br>The working range of the pressure instruments shall be chosen so that the maximum value is 75% of the range and the minimum is 15% of the range.<br>5) The accuracy class shall be 0.5 FS (full scale), or better.<br>-The total uncertainty of the instrument shall not exceed 0.5%, including process influences (temperature and static pressure) and combined effects of hysteresis and repeatability;<br>- The dead band shall not exceed 0.25% of the measuring range;<br>- A change in ambient temperature of up to 50ºC shall not affect the output signal by more than 1% of the measuring range.<br>6) Required for 2 valves or 5 valves<br>7) Min IP 56 / General Ex d<br>8) Required<br>9) Hart<br>10) Yes<br>11) Required (tray cover, thermal insulation)<br>12) Ex d ( if Ex ia is not required for specific application on the SPEC.. Ex. Zone 0)<br>13) Required for diaphram seal type<br>14) 1/2" NPT | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser. | NO FILED INSTRUMENTS IN OUR PACKAGE<br>21/07/2021: No pressure transmitter in our package, consequently, not applicable<br>HHI2021071 6] Noted. | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| 9 | Field Instrument | 3 | Local indicators (gauges)<br>1) Material (wetted part Vs non wetted part)<br>2) Cablibration range<br>3) Accuracy<br>4 Instrument manifold if applicable<br>5) IP<br>6) Glass size / type / Filled liquid / dial<br>7) Over pressure protector, over temperature protector, snubber(dampener)<br>8) Capillary tube protection if applicable<br>9) Flush ring with nipple & valve if applicable.<br>10) Measuring scale to be provided for all indicator as per project unit. | 1) AISI 316 stainless steel<br>2) Adequate for operating with working pressure equal to the upper limit of the nominal range<br>between 1/4 and 3/4 of the nominal range<br>3) ± 1,0% (full scale)<br>4) 3-way manifold shall be provided for impulse line installation or alternately, close-coupled AISI 316 stainless steel 2-valve manifold according API 551.<br>5) Min IP56<br>6) a nominal diameter of 114 mm or 100 mm<br>7) Yes if required<br>8) Yes<br>9) Yes for diaphram seal type<br>10) Double scale indication (kPa and bar) for pressure gauge | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser. | NO FILED INSTRUMENTS IN OUR PACKAGE<br>14/07/2021: To be discussed in the meeting<br>HHI20210716] Noted. However, Vendor to follow MR requirement for the instruments within vendor package | Closed |
| | | 4 | Flow meters<br>1) SIL<br>2) Material (wetted part Vs non wetted part)<br>3) Cablibration range/ Flow calculation / Turndown ratio<br>4) Accuracy<br>5) Instrument manifold if applicable<br>6) IP/ Ex. Proof.<br>7) Local indicator<br>8) Protocol ( Hart, FF, PB..)<br>9) Asset management interface configulation<br>10) Upstream / downstream straight length requirement.<br>11) Installation manual / recommendation from manufacturer.<br>12) Capillary tube protection if applicable<br>13) IS/Non IS<br>14) External power requirement<br>15) Flush ring with nipple & valve if applicable.<br>16) Cable Entry | 1) SIL 2 for Safety (Bidding)<br>2) Min SS316L for wettted part, In cases where the process is unsuitable for SS316L, higher grade material.<br>3) Min 10% higher than high alarm/ trip set point / Min 10% lower than low alarm / Trip set point<br>4) Refer to I-ET-3010.1M-1200-800-P4X-005 & I-ET-3010.00-1200-800-P4X-013<br>5) 5 valves for DP flowmeter<br>6) IP56 / General Ex d<br>7) Required<br>8) i) pulses up to 10kHz;<br>ii) totalized in the instrument transmitter itself and transmitted via non-proprietary field network.<br>9) Yes<br>10) As per manufacturer recommendation<br>11) To comply manufacturer recommendation<br>12) Required for diaphram seal type<br>13) Ex d ( if Ex ia is not required for specific application on the SPEC.. Ex. Zone 0)<br>14) Vendor to specify<br>15) Yes for diaphram seal type<br>16) NPT | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser. | NO FILED INSTRUMENTS IN OUR PACKAGE<br>21/07/2021: No flow meters in our package, consequently, not applicable<br>HHI20210716] Noted. | Closed |
| | | 5 | Level transmitter including chamber<br>1) SIL<br>2) Material (wetted part Vs non wetted part)<br>3) Cablibration range<br>4) Accuracy<br>5) Instrument manifold if applicable<br>6) IP/ Ex. Proof.<br>7) Local indicator<br>8) Protocol ( Hart, FF, PB..)<br>9) Asset management interface configulation<br>10) Installation manual / recommendation from manufacturer.<br>11) Capillary tube protection if applicable<br>12) Glass size / type<br>13) Vent/Drain valve for chamber<br>14) Flush ring with nipple & valve if applicable. | 1) SIL 2 for Safety (Bidding)<br>2) Min SS316L for wettted part, In cases where the process is unsuitable for SS316L, higher grade material.<br>3) Min 10% higher than high alarm/ trip set point / Min 10% lower than low alarm / Trip set point<br>4) Refer to I-ET-3010.1M-1200-800-P4X-005 & I-ET-3010.00-1200-800-P4X-013<br>5) 5 valves for DP flowmeter<br>6) IP56 / General Ex d<br>7) Required<br>8) Hart.<br>9) Yes<br>10) To comply manufacturer recommendation<br>11) Required for diaphram seal type<br>12) Use glass sections of dimensions 7, 8 or 9 with a maximum height limited to 5 (five) units of size 9<br>13) Level gauges shall have drain and vent connections with valves and caps included in the instrument<br>14) Yes for diaphram seal type | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser. | NO FILED INSTRUMENTS IN OUR PACKAGE<br>14/07/2021: No llevel transmitters in our package, consequently, not applicanble<br>HHI20210716] Noted. | Closed |
| | | 6 | Solenoid valves<br>1) Rating<br>2) IS/Non IS<br>3) Dual solenoid requirement | 1) Limited to 5W<br>2) NON IS ( Exd)<br>3) As per P&ID legned. | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser. | NO FILED INSTRUMENTS IN OUR PACKAGE<br>HHI20210716] Noted. However, Vendor to follow MR requirement for the instruments within vendor package | Closed |
| | | 7 | Temperature transmitter/Gauge with Termowell<br>1) SIL<br>2) Material (wetted part Vs non wetted part)<br>3) Cablibration range<br>4) Accuracy<br>5) Glass size / type<br>6) IP/ Ex. Proof.<br>7) Local indicator<br>8) Protocol ( Hart, FF, PB..)<br>9) Asset management interface configulation<br>10) Mininum distance requirement between temp. gauges and temp. transmitters.<br>11) Installation manual / recommendation from manufacturer.<br>12) Stress calculation<br>13) Thermowell | 1) SIL 2 for Safety (Bidding)<br>2) Min SS316L for wettted part, In cases where the process is unsuitable for SS316L, higher grade material.<br>3) Min 10% higher than high alarm/ trip set point / Min 10% lower than low alarm / Trip set point<br>The indication range shall be such that the normal operating temperature is between 30 and 70% of this range, and the minimum and maximum temp. ( For temp. Gauge)<br>4) ± 0.5 °C.<br>5) Display with minimum nominal diameter of 100 mm for gauge<br>6) IP56 / General Ex d<br>7) Required<br>8) Hart<br>9) Yes<br>10) As per P&ID requirement if applicable<br>11) To comply manufacturer recommendation<br>12) To be provided<br>13) For flanged wells, materials and construction shall respect the following order to maintain compatibility with connection to piping or equipment: welded to flange with full penetration weld in accordance with API RP 551 and ASME PTC 19.3 TW, the material of flange and the thermowell being AISI 316L stainless steel. When the AISI 316L stainless steel flange is not suitable for process conditions or line specification, the thermowell shall be made from a single AISI 316 stainless steel piece, without any welds. If this is still not suitable for process conditions, another material, equal to flange and body, with corrosion resistance and superior mechanical strength such as Duplex and Superduplex shall be selected. | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Bidder to recheck. There are field instruments such as pressure gauge, level sensor, oil in water analyser. | NO FILED INSTRUMENTS IN OUR PACKAGE<br>21/07/2021: no temperature transmitter in our package, consequently, not applicable<br>HHI20210716] Noted. | Closed |
| 10 | FGS detector | 1 | Fusible lines / Heat sensitive cable<br>1. If fusible line/Heat sensitive cable is required for PKG skid, fusible lines/Heat sensitive cables shall be supplied/installed by PKG vendor.<br>2) If fusible lines/Heat sensitive cable can't be installed by vendor, technical deviation shall be submitted for approval.<br>3. When fusible lines/Heat sensitive cable is not supplied and installed by vendor based on agreement, the support of fusible line/Heat sensitive cable shall be supplied on skid by PKG vendor to avoid hot work damage at yard.<br>Ex) Fusible line is applied to vessels, PKG vendor shall install required support on the vessel skid | The Fusible Plug system shall conform to DR-ENGP-M-I-1.3 – SAFETY ENGINEERING.<br>• High-risk areas are protected by fusible plugs or by fusible plugs and flame detectors, which, in presence of fire, bleed the deluge valve actuator and flood the area with seawater. For further information, refer to I-ET-3010.1M-1200-800-P4X-005 - FIELD INSTRUMENTATION and I-FD-3010.1M-5400-947-P4X-001 - SAFETY DATA SHEET. | I-ET-3010.00-1200-800-P4X-013<br>DR-ENGP-M-I-1.3<br>I-MD-3010.1M-5520-800-P4X-001 | | NO | Noted. | NO F&G DETECTORS IN OUR PACKAGE | Closed |
| | | 2 | VESDA(Very Early Smoke Detection Apparatus)<br>1. If VESDA is required for electrical panel supplied by PKG vendor, VESDA pipe support within panel shall be supplied and installed by PKG vendor. | N/A | N/A | | NO | Noted. | NO F&G DETECTORS IN OUR PACKAGE | Closed |
| | | 3 | Flame detectors<br>1. Flame detectors type shall be selected as per project requirement.<br>2) Calibration tools shall be supplied by vendor. | 1) Multi-Spectrum type (IR3) detectors with microprocessor technologies<br>Weather protection shall be provided for all detectors installed in open area.<br>2) A calibration kit, one per detector type, shall be provided | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Noted. | NO F&G DETECTORS IN OUR PACKAGE | Closed |
| | | 4 | Smoke detectors<br>1. Smoke detectors type shall be selected as per project requirement.<br>2) Calibration tools/tubing shall be supplied by vendor.<br>3.For line monitoring purpose, required resistance and diode shall be supplied/installed by PKG vendor. The value of resistance/diode will be informed during detailed engineering stage. | Usiung smoke detectors shall be reviewed before application for PKG.<br>1) Point optical detectors shall be used for smoke detection.<br>2) A calibration kit shall be provided, one per detector type | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Noted. | NO F&G DETECTORS IN OUR PACKAGE | Closed |
| | | 5 | IR point gas detectors<br>1. IR point gas detectors type shall be selected as per project requirement.<br>2) Calibration tools/tubing shall be supplied by vendor. | 1) Combustible gas detectors (point or open path type IR) shall be used to monitor flammable gas (CH4) leaks.<br>2) A calibration kit shall be provided, one per detector type | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Noted. | NO F&G DETECTORS IN OUR PACKAGE | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| | | 6 | Heat detectors<br>1. Heat detectors type shall be selected as per project requirement.<br>2) Calibration tools/tubing shall be supplied by vendor.<br>3. For line monitoring purpose, required resistance and diode shall be supplied/installed by PKG vendor.<br>The value of resistance/diode will be informed during detailed engineering stage. | Usiung heat detectors shall be reviewed before application for PKG.<br>1) Heat detectors shall be "raise of temperature rate" (Thermovelocimetric) type or fixed temperature type according to their location<br>2) A calibration kit shall be provided, one per detector type | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-1200-800-P4X-013 | | NO | Noted. | NO F&G DETECTORS IN OUR PACKAGE | Closed |
| | | 7 | Shutdown manual stations(push button)/ manual call points<br>1. Shutdown manual stations(push button)/ manual call points type shall be selected as per project requirement.<br>2)For line monitoring purpose, required resistance and diode shall be supplied/installed by PKG vendor. The value of resistance/diode will be informed during detailed engineering stage. | 5.5.3.10 There are various AMIs, also called Manual Call Points installed in suitable zone locations, around the field. Basically, these devices switch an electric contact upon the intentional lifting of their lid and push button manual actuation. Normally, the AMI switches are part of the AFDS but, in special cases, some AMI may be connected to a digital input card of a Remote I/O Panel of the FGS System. In this case, the connection shall have line monitoring.<br>5.3.5.1 Field ESD shutdown manual push-buttons shall have latching action and require local reset.<br>7.9.1 If required by Classification Society, MANUAL ESD STATIONS shall be provided in open areas, and their quantities and locations shall be defined during the Project's Detail Engineering Design Phase. | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-5140-700-P4X-002<br>I-MD-3010.1M-5520-800-P4X-001<br>I-ET-3010.00-5520-861-P4X-001 | | NO | Noted. | NO F&G DETECTORS IN OUR PACKAGE | Closed |
| | | 8 | Beacon / horn<br>1.Beacon / horn type shall be selected as per project requirement.<br>2)For line monitoring purpose, required resistance and diode shall be supplied/installed by PKG vendor. The value of resistance/diode will be informed during detailed engineering stage. | Beacon / horn<br>1.Beacon / horn type shall be selected as per project requirement.<br>2)For line monitoring purpose in cases of non fail safe, required resistance and diode shall be supplied/installed by PKG vendor. The value of resistance/diode will be informed during detailed engineering stage. | I-ET-3010.1M-5510-767-PPT-001 | | NO | Noted. | NO F&G DETECTORS IN OUR PACKAGE | Closed |
| 11 | Cable | 1 | Cable type(fire resistant, flame retardant…), color and etc shall meet project cable specification. | 1. For Electrical Cables<br>- Refer to clause 5.13 of I-ET-3010.00-5140-700-P4X-002 SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS.<br>- Armoured cable shall be applied except the wires in the panels.<br>- For cables to be installed only inside the accommodation areas, all materials used on their formation shall be halogen free.<br>- All cables shall be at least flame retardant.<br>- Cables for circuits that shall operate under fire conditions (systems for firefighting, fire and gas detection, alarm, public address, shut-down, emergency switch-off, emergency lighting, etc.), cables that feed essential and emergency services that are installed in hazardous areas, and cables crossing machinery space category A, as defined by SOLAS, shall be certified for circuit integrity under fire conditions, according to IEC 60331. These cables and related brackets may also be painted with specific products intended to keep their integrity for the expected fire and time conditions. The test reports of these products and the application procedures shall be subjected to PETROBRAS approval.<br><br>2. For Automation and Instrument cables<br>- Refer to clause 16 of I-ET-3010.00-1200-800-P4X-013 - GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS.<br>- Cable for intrinsically safe circuits shall be armoured. (Both indoor and outdoor) | I-ET-3010.00-5140-700-P4X-002<br>I-ET-3010.00-1200-800-P4X-013 | YES | | Noted | | Closed |
| | | 2 | All cable shall be binded with cable tie as per project requirement | All cable shall be binded with cable tie and/or cable cleat as per project requirement | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 3 | Cable coils/ loops for field devices shall be provided to avoid water dripping and fucture location adjusting. | Cable coils/ loops for field devices shall be provided to avoid water dripping and fucture location adjusting. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE, NO FIELD DEVICES | Closed |
| | | 4 | Cables shall laid on cable tray. Cables running without cable tray shall be approved before implementation. ( For whole skid including edge interface.) | Cables shall laid on cable tray. Cables running without cable tray shall be approved before implementation. ( For whole skid including edge interface.)<br>It is not permitted to install cables in flat bars;<br>Armoured cables shall be installed in cable trays or perforated channels; | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 5 | Wire color shall be in line with project requirement. | Cable core color shall be in line with project requirement which to be advised during detailed engineering. | For instrument, I-ET-3010.00-1200-800-P4X-013<br>Refer to attachment#3 | YES | | Noted | | Closed |
| | | 6 | Armour material shall comply to project requirement. | Galvanized steel threads braid protected by anti-corrosive cover for multi-core signal or power cables.<br>Copper or other non-magnetic metal threads braid protected by anti-corrosive cover for single-core cables. | I-ET-3010.00-5140-700-P4X-002 | YES | | Bidder to comply with I-ET-3010.00-5140-700-P4X-002. | ACCORDING TO OUR STANDARD CABLES<br>14/07/2021: noted | Closed |
| | | 7 | If fiber optic cable is required, the selection of multimode VS singemode shall be approved during detail engineering based on project specification. | If fiber optic cable is required, the selection of multimode VS singemode shall be approved during detail engineering based on project specification. | N/A (HHI Standard) | YES | | Noted | IF APPLICABLE | Closed |
| | | 8 | Minimum of spare capacity shall be provided in multicore cables as per project requirement. | Multi pair / triple / core cables shall have at least 20% spare at Mechanical Acceptance. | TBD | YES | | Noted | | Closed |
| | | 9 | Cable Code List | Enforce that CONTRACTOR's sub-suppliers and PACKAGED UNIT's SUPPLIERS also create such codes and emit a similar document. | I-ET-3010.00-1200-800-P4X-010<br>Refer to Form-11 | YES | | Noted | | Closed |
| | Lighting & Small Power | 1 | Where specified on the Purchaser's Electrical Requirements for Packaged Equipment Data Sheet, the Supplier shall provide lighting fixtures and/ or socket outlets.<br>Lighting fixtures and socket outlets when required shall be provided in accordance with project Lighting and Small Power Systems requirement. | Lighting fixtures and socket outlets to be supplied shall fulfill the requirement specified in I-ET-3010.00-5140-700-P4X-002 SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS.<br>And lighting and small power incl. socket outlet shall be designed in accordance with I-ET-3010.00-5140-700-P4X-001 SPECIFICATION FOR ELECTRICAL DESIGN FOR OFFSHORE UNITS. | I-ET-3010.00-5140-700-P4X-002<br>I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 2 | Lighting type shall be selected as per project requirement depending on application.<br>Ex) Indoor, Outdoor areas and inside of cabinet. | 1. Ligihting type shall be selected as per project requirement depending on application.<br>Ex) Indoor, Outdoor areas and inside of cabinet.<br>2. Extra safeguarding against falling down if the screwed connections loosen, shall be provided, refer to I-DE-3010.00-5140-700-P4X-001_LIGHTING INSTALLATION TYPICAL DETAILS.<br>3. Floodlight and searchlight installed at hazardous areas or open areas shall have plug-socket set for removal and maintenance at floor level. These plug-socket sets shall be Ex-e type according hazardous area classification. | I-ET-3010.00-5140-700-P4X-002<br>I-DE-3010.00-5140-700-P4X-001 | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 3 | A maintenance factor shall be used in design calculations according to project requirement.<br>and the lumen output of lamps shall be "average through life" value. LED shall have dispersive metal reflectors and shall be fitted with anti-vibration mountings. | N/A | N/A | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 4 | Lighting fixtures shall be solidly fixed and not suspended by means of chains or conduits, etc. with the exception of stanchion mounted fixtures. They shall be mounted such that routine maintenance can be carried out safely and without the use of temporary scaffolding. | Lighting fixtures shall be solidly fixed and not suspended by means of chains or conduits, etc. with the exception of stanchion mounted fixtures. They shall be mounted such that routine maintenance can be carried out safely and without the use of temporary scaffolding. | N/A (HHI Standard) | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 5 | Light fixtures shall be mounted at sufficient elevation and directed so as to avoid stroboscopic effects. | Light fixtures shall be mounted at sufficient elevation and directed so as to avoid stroboscopic effects. | N/A (HHI Standard) | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 6 | A maximum of 12 lighting fixtures per lighting loop shall be applied. The arrangement shall be such that failure of one lighting loop will not result in complete darkness of one area. Therefore as a minimum two lighting loops per area shall be applied. | The lighting circuits shall be distributed in order not to exceed 16A or to feed no more than 24 lighting points (each lamp is deemed to be a point, while each convenience outlet comprises 2 points).<br>A minimum of 2 (two) lighting circuits shall be provided on each area or compartment, being one normal and one essential. | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 7 | Lighting equipment shall be installed in such a manner that it will not obstruct or impair the clearance of passageways for personnel or equipment and such that shadows are not cast on valves, controllers, control panels, etc | Lighting equipment shall be installed in such a manner that it will not obstruct or impair the clearance of passageways for personnel or equipment and such that shadows are not cast on valves, controllers, control panels, etc | N/A (HHI Standard) | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 8 | The maximum number of convenience outlets on one supply shall comply to project requirement | The lighting circuits shall be distributed in order not to exceed 16A or to feed no more than 24 lighting points (each lamp is deemed to be a point, while each convenience outlet comprises 2 points). | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 9 | All maintenance sockets circuits shall not be mixed with lighting circuits if it is not allowd on project specification. | Socket-outlets and lamps shall be fed by different circuits. | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 10 | Socket/ plug type shall comply with project requirement. And Male plug also to be provided in case of outdoor and indurstical type of socket. | Socket/ plug type shall comply with project requirement, refer to clause 5.7 of I-ET-3010.00-5140-700-P4X-002.<br>And Male plug also to be provided in case of outdoor and indurstical type of socket. | I-ET-3010.00-5140-700-P4X-002 | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 11 | Interface Junction box for skid lighitng and socket to be provided by VENDOR.<br>Wiring DWG/layout for above to be provided. | Interface Junction box for skid lighitng and socket to be provided by VENDOR.<br>Wiring DWG/layout for above to be provided. | I-ET-3010.00-5140-700-P4X-003 | | NO | Noted | LIGHTING ARE NOT INCLUDED IN OUR SCOPE | Closed |
| | | 1 | The VENDOR shall provide all the required control stations such as push buttons, selector switches, indicating lamps, to be mounted on the package as per project requirement. | The VENDOR shall provide all the required control stations such as push buttons, selector switches, indicating lamps, to be mounted on the package as per project requirement. | N/A (HHI Standard) | | | Bidder to check | 14/07/2021: noted | Closed |
| | | 2 | The local control stations shall be installed on suitable steel supports near the respective pump/skids. The motor tag number and description shall be clearly identified close to the start/stop button. The contact block shall be hermeticaly sealed individually explosion protected single pole units with 1 NO or 1NC spare contacts.<br>The location should be such that it would not hinder access for normal maintenance and would not be liable to impact damage. | The local control stations shall be installed on suitable steel supports near the respective pump/skids. The motor tag number and description shall be clearly identified close to the start/stop button. The contact block shall be hermetically sealed individually explosion protected single pole units with 1 NO or 1NC spare contacts. The location should be such that it would not hinder access for normal maintenance and would not be liable to impact damage. | N/A (HHI Standard) | YES | | Noted | | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) | | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Yes | No | | | |
| 13 | Local Control Station (Push-button) | 3 | The local control station shall be/have:<br>- Minimum IP rating as per project requirement.<br>- Explosion proof as per project requirement. In addition, it is resistant to saliferous weather conditions, corrosion and ambient environment as specified.<br>- 'Start' push button: color, contact type(NO/NC), button type to be in line with project requirement.<br>- 'Stop' push button: color, contact type(NO/NC), button type to be in line with project requirement.<br>- Protection cover for Emergenty push button if required.<br>- Bottom cable entry with ISO metric threads and provision for external earthing. Provision for two-cable entry and the spare entry shall be closed with dummy stopper plug. All unused entries shall have Ex'd' certified nickelplated brass plug to meet the requirement certification.<br>- Sunshade for LCS<br>- Label with LCS tag number, motor tag and description<br>- LCS shall be provided from nominated vendor. Nominated vendor will be confirmed during detail engineering. | The local control station shall be/have:<br><br>1. Control boxes for push-buttons and signalling installed in hazardous areas and for external equipment kept in operation during ESD condition, shall be Ex de type.<br>2. Control boxes for push-buttons and signalling shall be made of non-metallic materials according to clause 5.2 of I-ET-3010.00-5140-700-P4X-002.<br>3. Push-buttons for ON (START) function shall be without release (return after push). Push-buttons for OFF (STOP) function shall be with release (retain after push).<br>4. Push-buttons for OFF (STOP) function shall have means for locking with padlock in OFF (STOP) positions'.<br>5. All field push-button shall have a load tag and a load identification or push-button function identification. Identification plate shall have be in black acrylic engraved with white letter for equipment installed indoors or in stainless steel AISI-316L for equipment installed outdoors.<br>6. Push-buttons for ON (START) function shall have a clear protective cover in order to avoid involuntary operation, as defined in NR-12.<br>7. Unless otherwise stated, all threaded joints shall be taper type, NPT with standardized tolerances, according to ASME B1.20.1. Cylindrical threaded joints shall be acceptable for lighting fixtures for fluorescent lamps, for control boxes for push-buttons and signalling, for floodlights and for power socket-outlets. Cylindrical threaded joints for other applications shall be submitted to PETROBRAS approval. For equipment installed in hazardous areas, the threads shall comply with the requirements of IEC 60079-0.<br>8. All screws, nuts and washers shall be made of bichromatized steel or AISI-316L stainless steel. | I-ET-3010.00-5140-700-P4X-002 | YES | | Noted | PACKAGE LOCATED IN SAFE AREA INDORS, SO Ex AND OUTDOOR REQUIRMENTS ARE NOT APPLICABLE | Closed |
| 14 | Electrical Heat tracing | 1 | Electrical heat tracing shall meet the requirements of project specification for Electric Heat Tracing | Electrical heat tracing shall meet the requirements of project specification for Electric Heat Tracing | N/A (HHI Standard) | | NO | Noted | NO HEAT TRACING IN OUR PACKAGE | Closed |
| | | 2 | Material from electrical heat tracing vendor selected in Project shall be proivided in order to ensure electrical heat tracing sytem control. | Material from electrical heat tracing vendor selected in Project shall be proivided in order to ensure electrical heat tracing sytem control. | N/A (HHI Standard) | | NO | Noted | NO HEAT TRACING IN OUR PACKAGE | Closed |
| | | 3 | Eletrical heat tracing equipment (heat tracing cable, Power kit, end kit etc) including power cable from inerface junction box installation shall be completed on PKG skid.<br>External Power interface Junction box shall arranged on boundray of skid for Field cable termination. | Eletrical heat tracing equipment (heat tracing cable, Power kit, end kit etc) including power cable from inerface junction box installation shall be completed on PKG skid.<br>External Power interface Junction box shall arranged on boundray of skid for Field cable termination. | N/A (HHI Standard) | | NO | Noted | NO HEAT TRACING IN OUR PACKAGE | Closed |
| | | 4 | Following document for Electrical heat tracing wiring shall be provided, but not limited to the following.<br>- Electrical heat tracing caluculation note<br>- Electrical heat tracing Installation detail dwg<br>- Wiring diagram<br>- Electrical heat tracing fulling pipe layout dwg | Following document for Electrical heat tracing wiring shall be provided, but not limited to the following.<br>- Electrical heat tracing caluculation note<br>- Electrical heat tracing Installation detail dwg<br>- Wiring diagram<br>- Electrical heat tracing fulling pipe layout dwg | N/A (HHI Standard) | | NO | Noted | NO HEAT TRACING IN OUR PACKAGE | Closed |
| | | 5 | Electrical heat tracing shall be pre-installed in FACTORY and relevent qualitty inspection recommended by Electrical heat tracing vendor inspection shall be completed before shipping. | Electrical heat tracing shall be pre-installed in FACTORY and relevent qualitty inspection recommended by Electrical heat tracing vendor inspection shall be completed before shipping. | N/A (HHI Standard) | | NO | Noted | NO HEAT TRACING IN OUR PACKAGE | Closed |
| | | 6 | Heat tarcing in package equipment shall be provided by package vendor and also all cabling works between junction box and heat tracing shall be provided by vendor and general arrangement & wiring diagram for above shall be provided separately. | Heat tarcing in package equipment shall be provided by package vendor and also all cabling works between junction box and heat tracing shall be provided by vendor and general arrangement & wiring diagram for above shall be provided separately. | N/A (HHI Standard) | | NO | Noted | NO HEAT TRACING IN OUR PACKAGE | Closed |
| 15 | Obstruction Lights | 1 | Aircraft warning lights shall be installed on all structures 40 meters and above,<br>designed and constructed in accordance with local aviation regulation and project requirement. | The aircraft obstruction warning signalling shall comply with the requirements of NORMAM 05, NORMAM 27 and RIPEAM 72 standards. It shall be composed by warning lighting fixtures in towers, cranes, rigs and helideck.<br>Where it is not possible to install aircraft obstruction lights on obstacles (e.g., flare tower), floodlights shall be used as an alternative solution. Floodlights shall be positioned in order to not dazzle the vision of the pilots during landings and takeoffs. Refer to I-ET-3010.00-5140-700-P4X-002 - SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS. | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | NO HEAT TRACING IN OUR PACKAGE | Closed |
| | | 2 | Helideck lighting and associated illuminated wind sock shall be as per local aviation authority requirement and ICAO, CAP 437 and CAAP71 standards. | The control panel of this system shall automatically control the signalling and lighting of helideck (helideck perimeter lighting fixtures, touch area lighting, windsock lighting and runway searchlights, status light, etc.).<br>When the helideck is unguarded, or the parameters provided by the Helideck Monitoring System (HMS) are out of safety limits, or the offshore unit is under gas leakage, the status light shall remain on, according to NORMAM 27.<br>This system shall be supplied with 220VAC from essential lighting transformers.<br>For further information, refer to I-ET-3010.00-5140-700-P4X-002 - SPECIFICATION FOR ELECTRICAL MATERIAL AND EQUIPMENT FOR OFFSHORE UNITS. | I-ET-3010.00-5140-700-P4X-001 | | NO | Noted | NO HEAT TRACING IN OUR PACKAGE | Closed |
| 16 | Multi Cable Transits (MCT) | 1 | Wherever required, the supplier shall provide MCT for power, control, lighting, and<br>earthing cables. The MCT insert modules shall allow reconstituting the tightness of the wall, floor or roof. | Wherever required, the supplier shall provide MCT for power, control, lighting, and earthing cables. The MCT insert modules shall allow reconstituting the tightness of the wall, floor or roof. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 2 | The MCT frame and insert modules shall be of fire resistant materials. The MCT frame material shall be min 316L stainless steel if not specified on porject spec. and welded directly on the floor, wall or roof. | The MCT frame and insert modules shall be of fire resistant materials. The MCT frame material shall be min 316L stainless steel if not specified on porject spec. and welded directly on the floor, wall or roof. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 3 | All bolts nuts, and accessories shall be 316L stainless steel if not specified on porject spec. and suitable for marine applications. | All bolts nuts, and accessories shall be 316L stainless steel if not specified on porject spec. and suitable for marine applications. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 4 | Non-magnetic stay-plates and MCT frame body are required within the transits for routing single core AC cables (in trefoil formation). | Non-magnetic stay-plates and MCT frame body are required within the transits for routing single core AC cables (in trefoil formation). | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 5 | The MCT shall have the minimum A60 fire rating. And the rating shall be suitable for fire protection rating as per project requirement. | The MCT shall have the minimum A60 fire rating. And the rating shall be suitable for fire protection rating as per project requirement. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 6 | MCTs shall be peel-off type (similar to Roxtec ®) and certified for use in the hazardous area by approved international authority. | MCTs shall be peel-off type (similar to Roxtec ®) and certified for use in the hazardous area by<br>approved international authority. | N/A (HHI Standard) | | NO | | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 7 | Adequate spare MCT blocks (not bank type) shall be kept for future use. | Each MCT shall have at least 20% of spare blocks. | I-ET-3010.00-5140-700-P4X-002 | YES | | Noted | | Closed |
| | | 8 | The MCT location shall be arranged on proper location without interference for cable penetration. | The MCT location shall be arranged on proper location without interference for cable penetration. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 1 | Supplier to proivde suitable termination point for field interface cable and dedicated terminal block (ferrul connection type) and bus bar connection. | Supplier to proivde suitable termination point for field interface cable and dedicated terminal block and bus bar connection.<br>Cables shall be grouped on a properly identified terminal block. Each block shall have at least 20% of reserve terminals.<br>Terminals shall be eye type. | I-ET-3010.00-5140-700-P4X-002 | YES | | Noted | | Closed |
| | | 2 | Interwirng between componet and terminal block shall be provided to avoid direct connection to componets such as MCBs, PCB card and realys etc.. | Interwirng between componet and terminal block shall be provided to avoid direct connection to componets such as MCBs, PCB card and realys etc.. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 3 | In case that Main power cable for incomming outgoing is connected to bus bar, to apply Bus bar hole size shall be as below;<br>- Cable core size 95sqmm and below : Bus bar hole 11mm for 10 Pi terminal lug hole.<br>- Cable core size above 95sqmm : Bus bar hole 14mm for 12 Pi terminal lug hole. | In case that Main power cable for incomming outgoing is connected to bus bar, to apply Bus bar hole size shall be as below;<br>- Cable core size 95sqmm and below : Bus bar hole 11mm for 10 Pi terminal lug hole.<br>- Cable core size above 95sqmm and below : Bus bar hole 14mm for 12 Pi terminal lug hole. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 4 | If Two lug hole type is mendatory, Two hole space shall be 44.5mm as standard. And Terminal lug, bolt, Nut and accessory suitable for Project requirement shall provied by vendor as equipped on bus bars. | Compression terminal lug, bolt, Nut and accessory suitable for Project requirement shall be provided by vendor as equipped on bus bars.<br>If Two hole lug type is required, two hole space shall be 44.5mm as standard. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 5 | Special termination bus bar type is not accepted in general. Vendor shall obtain approval from Contractor to apply special type connection. In this case all termination material and accessary shall be provided Vendor as per project cable sizing and specification. | Special termination bus bar type is not accepted in general. Vendor shall obtain approval from Contractor to apply special type connection. In this case all termination material and accessary shall be provided Vendor as per project cable sizing and specification. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 6 | If terminal block naming procedure is not specifed on project requirement,<br>terminal block name shall follow below requireement for field interface cable.<br>- AC power line : XAC<br>- DC power line : XDC<br>- Control Signal Input: XCI<br>- Control Signal output: XCO | If terminal block naming procedure is not specifed on project requirement, terminal block name shall follow below requirement for field interface cable.<br>- AC power line : XAC<br>- DC power line : XDC<br>- Control Signal Input: XCI (for electric)<br>- Control Signal output: XCO (for electric)<br>For Instrument control<br>• The terminal blocks shall be segregated by each type of electrical signal, i.e., discrete input, discrete output, analog input, analog output, power supply, in order to assure proper segregation of signals.<br>• Numbering procedure shall be approved during detailed engineering | N/A (HHI Standard) | YES | | Noted | | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| 17 | Termination | 7 | In case of fiber optic cable termination as field cable, Vendor shall comply below requirement<br>- Dedicate cable route for Fiber optic cable in panel or equipment inside shall be provided, ensuring adequate cable bending radius to avoid damage of cable. and this route shall be clearly indicated in equipment layout vendor data.<br>- Patch box shall be equipped is panel (not ship loose) and to ensure Clear entrance for Fiber optic cable without obstruction.<br>- Patch box to have all termination material (cable gland or fitting for Fiber optic cable, splicing kit, pig tail etc.)<br>- Patch cord cable between Patch box and equipment shall be vendor scope and pre-wiring shall be completed in vendor factory.<br>- Manufacturer test report for internal Patch cord shall be submitted accordinlgy.<br>- Patch cord connect type shall be standard type as per project requirement. | In case of fiber optic cable termination as field cable, Vendor shall comply below requirement.<br>- Dedicate cable route for Fiber optic cable in panel or equipment inside shall be provided, ensuring adequate cable bending radius to avoid damage of cable. and this route shall be clearly indicated in equipment layout vendor data.<br>- Patch box shall be equipped is panel (not ship loose) and to ensure Clear entrance for Fiber optic cable without obstruction.<br>- Patch box to have all termination material (cable gland or fitting for Fiber optic cable, splicing kit, pig tail etc.)<br>- Patch cord cable between Patch box and equipment shall be vendor scope and pre-wiring shall be completed in vendor factory.<br>- Manufacturer test report for internal Patch cord shall be submitted accordinlgy.<br>- Patch cord connect type shall be standard type as per project requirement.<br>- Test report for patch cord / splicing to be supplied. | N/A (HHI Standard) | YES | | Noted | IF APPLICABLE FO OUR PACKAGE | Closed |
| | | 8 | In case of Patch cord cable termination as field cable, Vendor shall comply below requirement<br>- Dedicate cable route for patch cord in panel or equipment inside shall be provided, ensuring adequate cable vending radius to avoid damage of cable. and this route shall be clearly indicated in equipment layout vendor data.<br>- Patch cord connect type shall be standard type as per project requirement. | In case of Patch cord cable termination as field cable, Vendor shall comply below requirement.<br>- Dedicate cable route for patch cord in panel or equipment inside shall be provided, ensuring adequate cable bending radius to avoid damage of cable. and this route shall be clearly indicated in equipment layout vendor data.<br>- Patch cord connect type shall be standard type as per project requirement. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 9 | All field cables are allowed to be terminated in only front access of equipment without dismantling any componet or equipment. | All field cables are allowed to be terminated in only front access of equipment without dismantling any componet or equipment. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| | | 10 | For motor termination, The cable entry size and entry direction for all motors, equipments and instruments shall be indicated on vendor drawing and suitable for HHI's cable data accordingly.<br>For motors rated certain level(based on project requirement) above, a space heater shall be used.<br>Separated cable termination box shall be provided for motor power and space heater circuit for all motors for certain level(based on project requirement) above. | For motor termination, The cable entry size and entry direction for all motors, equipments and instruments shall be indicated on vendor drawing and suitable for HHI's cable data accordingly. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 11 | Size and quantity of terminal block or bus bar shall be suitable for HHI cable size and core Quantity.<br>Especially for bus bar connetion, bus bar hole size and dimension shall be compatible with HHI terminal lug size. All relevant data of cable will be informed later. | Size and quantity of terminal block or bus bar shall be suitable for HHI cable size and core Quantity.<br>Especially for bus bar connetion, bus bar hole size and dimension shall be compatible with HHI terminal lug size. All relevant data of cable will be informed later. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 12 | The terminaltion space between enclosure and terminal block or bus bar shall be sufficient for external cable connection. In case of cabinet, gap between power cable connetion point and bottom gland plate shall be over 300mm for adequate cable termination. | The terminaltion space between enclosure and terminal block or bus bar shall be sufficient for external cable connection. In case of cabinet, gap between power cable connetion point and bottom gland plate shall be over 300mm for adequate cable termination. | N/A (HHI Standard) | YES | | Noted | | Closed |
| | | 13 | 1. Terminal block material color for each signal type shall be inline with project requirement.<br>2) Terminal block and bus bar size shall be suitable for external cable size and q'ty which will be informed later<br>3. Terminal block for spare wires of multi cables shall be provided by Package vendor. For other methods such as floating of spare wires, it shall be approved via official documentation.<br>4. Terminal block type for each signal type shall be selected as per project requirement.<br>Ex. knife type, fused type, cage clamp type..... | 1. Electrical :<br>Unless otherwise stated in specific equipment technical specification, for control cables, these connectors shall be eye type.<br>Terminal lugs shall be suitable for naval use, shall be anti-vibration type and assembled on support profiles ("C" channels).<br>In order that neither destruction nor deformation of wires forming the cable occurs, terminal lugs shall be of indirect press over the conductor.<br>They shall have a minimum capacity of 20A/600V and shall be made of steatite or melamine insulation, not containing toxic or organic substances.<br>It shall not be accepted more than one cable connected to each terminal. In case of necessity of connection of more than one cable at the same point, it shall be used one terminal lug for each cable and these terminal lugs shall be connected by metallic bridge bars.<br>Jumpers between terminals through external conductors shall not be accepted in terminal blocks. For this purpose metallic bridge bars shall be used.<br>The terminals strips shall be installed in order to guarantee enough space to perform the cable terminations, easy access to terminals and easy reading of the identification rings.<br>The terminals strips shall be numbered with progressive numbers and codified as per electrical diagram indications.<br><br>2. Instrument<br>For external Wiring in control panel, Terminal connectors shall be<br>• Made of non-hygroscope material;<br>• Adequate for conductors with 2.5 and 1.5 mm 2 cross section;<br>• Flame retardant;<br>Insulated to 300 V;<br>• Insulated to 300 V;<br>• Of disconnect type (knife type) for external wiring / incoming cable<br>For Juction box<br>Signals to sub-systems, must be segregated in different terminal strips and multicables (where applicable), when installed inside the same junction box. The junction boxes shall be segregated into 3 types:<br>• TYPE 1: for signals using flame-resistant cables<br>• TYPE 2: for signals related to safety interlocks<br>• TYPE 3: for signals related to process control<br>3. Teminal block Color for instrument<br>• intrinsically safe: Blue<br>• Non intrinsically safe: Hold ( To be approved by HHI during detailed engineering)<br>4. Duct Color for instrument<br>• intrinsically safe: Blue<br>• Non intrinsically safe: Gray<br>5. Spare terminal block for control panel<br>20% wired etra terminal block and 10% for future use<br>Note: all cable screen, shielding shall be individually connected to terminal block, | I-ET-3010.00-5140-700-P4X-002<br>I-ET-3010.00-1200-800-P4X-002<br>I-ET-3010.00-5520-888-P4X-001 | YES | | Noted | | Closed |
| **3. UCP/LCP, DCS/SDS/FGS interface and requirement** | | | | | | | | | | |
| 1 | **IO segregation** | 1 | IO segregation shall be provided as per project requirement. (Ex. IS/NIS, Safety, FGS, Analogue/digital, subsystem..) | Package Units shall also follow the requirements established for the whole Automation and Control system, with the exceptions foreseen in the aforementioned specification. Control/ESD(Safety) shall be independent systems only where required by Classification Society. | I-MD-3010.1M-1200-940-P4X-025<br>I-ET-3010.00-1200-800-P4X-002<br>I-ET-3010.00-5520-862-P4X-001 | YES | | Noted | | Closed |
| 2 | **Voting / Degrading Redundancy** | 1 | Voting, degrading, redundancy shall be provided as per project requirement.<br>Below Is item list to check but not limited to:<br>1. IO cards/modules<br>2) Controllers/PLCs<br>3. Program and data memory in a PLC<br>4. Power supplys<br>5. Communication modules | General: Package Units shall also follow the requirements established for the whole Automation and Control system, with the exceptions foreseen in the aforementioned specification.<br>1. IO cards ( Bidding for redudant, In addition to provide cost saving for single )<br>2. The PLC CPU racks shall be arranged in redundant half-clusters in hot standby configuration.(In addition to provide cost saving for single )<br>3. The programming software shall allow visualization, monitoring, modifications, simulation of the application software from the programming tools (PC based) in both on-line and off-line modes.<br>Programming of Package Control System (controllers, HMI, switches, screens and all programmable/configured variables) as well as applicable software shall be supplied fully compatible with Microsoft Windows® based operating system.<br>4. Power supply shall comply with I-ET-3010.00-5140-700-P4X-003 - ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS.<br>5. Redundant Gigabit Ethernet (IEEE 802.3an) link shall be provided between Packages UCP and SOS Package Units Data Server.<br>Package. It is part of the panel SUPPLIER's scope to convert and redundantly distribute the different power supplies inside panels, including, where necessary, redundant AC/DC or DC/DC stabilized power supply units for cabinet's internal distribution of 24 VDC. et.<br>5. VENDOR to comply all requirement onl-MD-3010.1M-1200-800-P4X-001<br>6. When possible, to standardize equipment, engineering, construction and assembly and commissioning activities the CONTRACTOR shall procure PACKAGES with same PLCs models, same HMIs models. For PACKAGE switches that interfacing with CSS/HCSS and SOS System, the MANUFACTURER shall be the same of SOS System.<br>For all P2, P2S, P2C and P2SC PACKAGE units, CONTRACTOR shall send the properly templates (OPC UA, Modbus TCP/IP and any other protocol developed during detailed design phase) to PACKAGER in order to standardize the memory map presentation and to facilitate the configuration and tests between PACKAGES and CSS/HCSS/SOS Systems.<br>PACKAGE Servers shall be redundant.<br><br>. | I-MD-3010.1M-1200-940-P4X-025<br>I-ET-3010.00-5520-862-P4X-001<br>I-ET-3010.00-5520-888-P4X-001<br>I-MD-3010.1M-1200-800-P4X-001<br>I-ET-3010.00-5520-800-P4X-001<br>I-ET-3010.00-5520-861-P4X-001<br>I-ET-3010.00-5520-861-P4X-001<br>I-ET-3010.00-5520-861-P4X-002<br>I-ET-3010.00-5520-862-P4X-001 | | NO | Vendor to provide control system with redundant power supply. | NO REDUNDANCY IS INCLUDED IN OUR PACKAGE<br>14/07/2021 : to be discussed in the meeting<br>HHI20210716] Noted. However, Vendor to follow MR requirement if ther is power supplier in vendor package. | Closed |
| 3 | **SIL level<br>Reliability and Availability** | 1 | For safety system, SIL certified components shall be selected as per project requirement.<br>Ex) SIL relay, SIL certified PLC/ Controller and etc | SIL certification to be provided for Safety PLC and safety relay if applicable. | I-ET-3010.00-5520-888-P4X-001 | | NO | noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 4 | **Wire duct requirement (Including spare and yard cable)** | 1 | 1. Wire duct color shall meet project requirement<br>2) Wire duct size shall be designed to accommodate whole yard cable. Duct sizing calculation report shall be provided for approval during detailed engineering. | 1) Blue plastic ducts for intrinsically safe field wiring, plastic gray ducts for Non intrinsically safe.<br>2) Maximum 60% of their width are used, considering all spare and future I/O | I-ET-3010.00-5520-888-P4X-001 | | NO | Vendor to provide | NOT APPLICABLE FOR OUR PACKAGE<br>14/07/2021: noted<br>HHI20210716] Noted. we will check detail design stage. | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) | | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Yes | No | | | |
| 5 | **Time synchronization** | 1 | Time synchronization shall be provided as per project requirement if applicable. | PACKAGE UNITs´ alarms acknowledgment applied from PACKAGE HMI or from the dedicated workstation shall be synchronized with UNIT SOS. PACKAGE SUPPLIER shall implement acknowledgement feature in PACKAGE UNIT side (controller), so that this acknowledgement can be easily readable from SOS PACKAGE UNIT RTDS.<br>Time synchronism for all A&C systems shall be sent from Time Servers (part of Electrical System), mounted inside Time Server Panel (PN-5140005), through SNTP protocol. This signal shall be sent to SOS LAN, CSS LAN and Package<br>Unit LAN, in order to be available for all A&C systems. For further information, see I-MD-3010.1M-5520-800-P4X-003 - AUTOMATION NETWORK DESCRIPTION.<br>All Automation equipment shall be able to synchronize their clocks using the SNTP packets circulating in the automation rings. Therefore, all Automation network equipment (switches L2 and L3) shall be SNTP compatible and replicate the signal and to make the Synchronization signal reach all workstations and panels connected to the network.<br>Item 5.2.10.2_I-MD-3010.1M-5520-800-P4X-003 is also valid for equipment inside package units, i.e., every package PLC, server and other equipment shall use the SNTP packets for their own synchronization. In order to do so, the switches inside the packages shall also be compatible with SNTP and able to propagate the SNTP packets to the package equipment. | I-ET-3010.00-1200-800-P4X-002<br>I-MD-3010.1M-5520-800-P4X-001<br>I-MD-3010.1M-5520-800-P4X-003 | | NO | Vendor to provide | NOT APPLICABLE FOR OUR PACKAGE<br>14/07/2021: to be discussed in the meeting<br>HHI[20210716] Noted. However, Vendor to follow MR requirement if needed. | Closed |
| 6 | **Watchdog** | 1 | Watchdog shall be provided as per project requirement if applicable. | Watchdog requirement to be informed in detailed engineering. ( If applicable) | I-ET-3010.00-1200-800-P4X-002 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 7 | **Control Panel FAT** | 1 | Factory Acceptance Test for control panel shall be conducted as per project requirement.<br>FAT procedure for control panel shall be submitted for approval. | FAT procedure for UCP shall be provided for review | I-MD-3010.1M-1200-800-P4X-001 | YES | | Noted | AS PER FAT PROTOCOL FOR THE PACKAGE | Closed |
| 8 | **Minimum hard IO interface for DCS** | 1 | Minimum hard IO interface shall be provided as per project requirement.<br>Ex) Common fault, time sync……and etc | Hardwire interface specified on I-ET-3010.1M-1200-800-P4X-014 shall be provided as minimum.<br>In addition, cabinet common alarm for earth fault, fan failure to be provided. | I-ET-3010.1M-1200-800-P4X-014 | YES | | Noted | | Closed |
| 9 | **ESD/SDS IO interface** | 1 | Any safety related interface between PKG UCP and ICSS(ESD/SDS) shall be hardwired IO. | Hardwire interface specified on I-ET-3010.1M-1200-800-P4X-014 shall be provided as minimum. | I-ET-3010.1M-1200-800-P4X-014 | YES | | Noted | | Closed |
| 10 | **FGS IO interface** | 1 | Any interface between PKG UCP and FGS shall be hardwired IO. | Hardwire interface specified on I-ET-3010.1M-1200-800-P4X-014 shall be provided as minimum. | I-ET-3010.1M-1200-800-P4X-014 | | NO | | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 11 | **Panel color / Size / type** | 1 | Panel color, size, type shall be inline with project requirement. | 1) Internal and external Automation panels color shall be light cream (Munsell notation 2.5 Y 9/4)<br>2) Maximum size of each section shall be 800 mm x 800 mm x 2,000 mm (length x width x height).<br>3) Side plates shall not be used between panel sections. Standardized beams and/or struts shall be used for section reinforcement. Doors shall be of single type and constructed with a 14 gauge (or thicker) steel plate, "U" beam reinforcement, grounding bolts, internal hinges, locks and handles. Locks, handles and hinges shall be designed to allow changing of door opening direction without any further piece of equipment.<br>4) Control Panel requirement specified in clause 4.8 of I-ET-3010.00-5140-700-P4X-002 shall be fulfilled. | I-ET-3010.00-5520-888-P4X-001 | YES | | Noted but bidder to comply with painting and color. | ACCORIDING TO OUR STANDARDS<br>14/07/2021: noted | Closed |
| 12 | **Line monitoring** | 1 | Line monitoring to be provided as per project requirement. If diode or resistantance shall be installed on field devices such as detectors for line monitoring, all field device shall be supplied after installing the required items. | For non fail safe, diode and resistor shall be provided for line monitoring.<br>For interface with CSS, detail SPEC of resistor and diode will be proivded by CONTRACTOR and VENDOR shall install them at panel and field devices accordingly | I-ET-3010.00-5520-888-P4X-001 | | NO | | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 13 | **Failure mode alarm** | 1 | Failure mode alarm shall be provided as per project requirement. | Failure mode alarm specifed on I-ET-3010.00-5520-888-P4X-001 and I-ET-3010.00-1200-800-P4X-002 shall be provided. | I-ET-3010.00-5520-888-P4X-001<br>I-ET-3010.00-1200-800-P4X-002 | | NO | | NOT APPLICABLE | Closed |
| 14 | **Firewall** | 1 | If applicable, Vendor to install the fire wall as per project requirement. | All firewall required for CSS interface shall be supplied by PKG VENDOR (IEC 62443) | IEC 62443 | | NO | | NOT APPLICABLE | Closed |
| 15 | **HMI/Laptop/Software license** | 1 | HMI/Laptop and software license shall be provided as per project requirement. | Programming of Package Control System (controllers, HMI, switches, screens and all programmable/configured variables) as well as applicable software shall be supplied fully compatible with Microsoft Windows® based operating system.<br>Package supervisory system software shall be supplied to be installed at Engineering Workstation. Package UCP controller software shall be supplied to be installed at Maintenance Workstation.<br>All keys, drivers, manuals and licenses of all software inside Package shall be provided. No software access restrictions will be accepted.<br>The programming software shall allow visualization, monitoring, modifications, simulation of the application software from the programming tools (PC based) in both on-line and off-line modes.<br>Supply for each PACKAGE the special tools, special cables, programmers, converters, licensed software, for allow configuration, maintenance and troubleshooting of PACKAGE equipment as PLCs, HMIs, switches, etc<br>Provide the following PACKAGE software on original specific media:<br>• PLC programming software;<br>• HMI system development software;<br>• The software of monitoring system;<br>• All process recipes of PLC and of HIM;<br>• Any other software necessary to reinstall all the control system inside the PACKAGE.<br>• All devices inside of PACKAGE with Microsoft operational system shall be provide with the software on original specific CD/DVD media and licenses. OEM licenses are not acceptable.<br>Before a Handover (after SIT) of PACKAGE, provide a bootable image of each computer in the PACKAGE.<br>The same software installed in PACKAGE Maintenance Workstations and PN-5500005A/B shall create computer images.<br>Supply for each PACKAGE the special tools, special cables, programmers,converters, licensed software, for allow configuration, maintenance and troubleshooting of PACKAGE equipment as PLCs, HMIs, switches, etc. | I-ET-3010.00-1200-800-P4X-002<br>I-MD-3010.1M-1200-800-P4X-001<br>I-MD-3010.1M-1200-800-P4X-001<br>I-ET-3010.00-5520-800-P4X-001<br>I-ET-3010.00-5520-861-P4X-001<br>I-ET-3010.00-5520-861-P4X-002<br>I-MD-3010.1M-5520-800-P4X-001<br>I-ET-3010.00-5520-862-P4X-001 | | NO | noted | THE SOTWARE OF OUR PACKAGE IS A COMMERCIAL SOFTWARE, WE DO DEVELOP AN SPECIFIC ONE FOR OUR PACKAGE. THEREFORE, NO LICENCES ARE SUPPLIED | Closed |
| 15 | **HMI/Laptop/Software license** | 1 | HMI/Laptop and software license shall be provided as per project requirement. | PAS HMI shall comply all requirement including Symbols and color codes for graphic on I-ET-3010.00-5520-800-P4X-001<br>The PACKAGE HMI screens shall cover all PACKAGE UNIT functionalities, including not only PACKAGE UNIT´s own process, but also its diagnostic information. Data and variables shall be displayed following PACKAGE UNIT Process and Instrumentation Diagrams (P&ID). The PACKAGE HMI screens shall be developed in a way that it is easy for the operator to understand its contents. Real-time values, equipment status changes, alarms and failures shall be highlighted and promptly identifiable. There shall be at least one screen with a general overview of the PACKAGE UNIT main variables.<br>HMI screens shall be in Brazilian Portuguese language and in additional languages requested in the PACKAGE UNIT technical specification<br>P2x PACKAGE UNITs HMIs and dedicated workstation(s) appearance and functionalities shall follow the defined in I-ET-3010.00-5520-800-P4X-001 – SUPERVISION AND OPERATION SYSTEM (SOS) SCREENS, whenever possible, to keep PACKAGE UNITS´ operation screens similar to SOS screens.<br>At least, the following items shall be complied: engineering units, controllers mode (direct/reverse), control valves´ indication, level indication, line colors and open/close/start/stop indications.<br>For P1 type Packages, the operator interface shall have at least:<br>• Push-buttons and leds signaling;<br>• Small electronic graphic units and keyboard;<br>• Video screen and keyboard.<br>For P2 type Packages, the operator interface shall be mounted at the external face of the panel front door. All supervision and control and start/stop functions, local data archiving, maintenance and configuration functions, alarm processing and status displays shall be provided at UCP HMI.<br>Maintenance Workstations (PN-5500005A/B) supplied by CSS shall contain all software needed to configure, program and test CSS controllers and package units´ controllers. All required support and SW for Maintenance Workstations to be supplied by PAS supplier. | I-ET-3010.00-1200-800-P4X-002<br>I-MD-3010.1M-1200-800-P4X-001<br>I-MD-3010.1M-1200-800-P4X-001<br>I-ET-3010.00-5520-800-P4X-001<br>I-ET-3010.00-5520-861-P4X-001<br>I-ET-3010.00-5520-861-P4X-002<br>I-MD-3010.1M-5520-800-P4X-001<br>I-ET-3010.00-5520-862-P4X-001 | | NO | noted | THE SOTWARE OF OUR PACKAGE IS A COMMERCIAL SOFTWARE, WE DO DEVELOP AN SPECIFIC ONE FOR OUR PACKAGE. THEREFORE, NO LICENCES ARE SUPPLIED | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) Yes | Comply (Vendor to fill out) No | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| 16 | **Cyber security** | 1 | Cyber security shall comply to project requirement.<br>Below Is item list to be checked but not limited to:<br>-Back-up and restore procedure shall be written and tested. (If required in project specification)<br>-External mass storage device(USB..) shall not be accessible in normal operation.<br>-Access level shall be configured as per project requirement. | 18.1.1 The Automation Network is described in I-MD-3010.1M-5520-800-P4X-003 – AUTOMATION NETWORK DESCRIPTION. Automation Network equipment specification is defined in I-ET-3010.1M-5520-800-P4X-004 – AUTOMATION NETWORK REQUIREMENTS.<br>18.1.2 All A&C systems shall be designed taking into account IEC 62443-2-1 - Industrial communication networks – Network and system security – Part 2-1: Establishing an industrial automation and control system security program and IEC 62443-3-3 - Industrial communication networks – Network and system security – Part 3-3: System security requirements and security levels.<br>18.1.3 Writing in any process variables externally from Automation Network shall not be allowed. Any external access to Automation Network shall be previously authorized and registered.<br>18.1.4 The connections from Automation Network to Automation firewall are from Package Unit LAN, CSS LAN and SOS LAN. Any system that shall communicate with Corporate Network, such as FMS, PI, SPCS, AMS and MMS shall one of these connections. Specific policies shall be foreseen in Automation Firewall for these communications.<br>Vendor to comply IEC 62443<br>18.1.5 Anti-virus, anti-spyware, anti-malware, access register/control, version control, USB ports blocking and other protection means shall be foreseen in all A&C systems.<br>18.1.6 All anti-virus/anti-malware/anti-spyware to be installed in SOS computers shall be certified by the Supervisory System manufacturer.<br>18.1.7 All Automation systems shall be supplied cleaned against anti-virus/anti-malware.<br>18.1.8 Software suppliers shall issue a report containing information on which Security Levels are achieved by their product.<br>* All firewall required for CSS interface shall be supplied by PKG VENDOR (IEC 62443)<br>* Temporary Antivirus<br>A temporary antivirus approved by the Automation System Vendor and compatible with Automation Supervisory and Controller systems shall be installed on all Automation system servers and workstations.<br>Licenses shall start at the beginning of the construction and assembly time period, and shall be renewed for as many times as necessary until the definitive antivirus is installed or until 3 years after the first oil event date (i.e., the actual event date, not its predictions). The license renewals during this time period, the eventual time schedule changes or first oil postponing shall not incur in additional costs to PETROBRAS.<br>VENDOR shall support all requirement for backup management during detailed engineering<br>PACKAGE MANUFACTURER shall issue a document presenting all applications configured, procedures and recommendations for application programs back-up and restore. | I-MD-3010.1M-5520-800-P4X-001<br>IEC 62443<br>HHI Standard<br>I-MD-3010.1M-5520-800-P4X-003<br>I-DE-3010.1M-5520-800-P4X-002 | | NO | vendor to provide | IN OUR OPINION, IT IS NOT APPLICABLE FOR OUR PACKAGE<br>14/07/2021: The bilge alarm in protected by a passwword, in order to avoid any changes in the configuration.<br>PLC is also protected by a password.<br>HHI20210716] Noted. | Closed |
| 17 | **Surge arrestor** | 1 | Surge arrestor shall be provided for required eletrical loop as per project requirement. | N/A | N/A | | | | N/A | Closed |
| 18 | **Accessaries** | 1 | Following panel accessaries shall be supplied by vendor according to project requirement.<br>1. Warming label (power circuit)<br>2) Protection cover (power circuit)<br>3. Anti vibration pad<br>4. Door lock / stopper (hinge)<br>5. Tag plate<br>6. Lighting/door switch<br>7. Power socket<br>8. Fan / Temperature sensor<br>9. Earthing monitoring system | Following panel accessaries shall be supplied by vendor according to project requirement.<br>1. Warming label (power circuit)<br>2) Protection cover (power circuit)<br>3. Anti vibration pad<br>4. Door lock / stopper (hinge)<br>5. Tag plate<br>6. Lighting/door switch<br>7. Power socket<br>8. Fan / Temperature sensor<br>9. Earthing monitoring system | I-ET-3010.00-5520-888-P4X-001 | YES | | Noted | IF REQUIRED AND APPLICABLE | Closed |
| 19 | **Space requirement** | 1 | 1. Terminal block, relay, IO card/module, IO module slot, CPU, space spare, and others for future shall be provided as per project requirement.<br>2) Calculated spare shall be specifed on UCP panel DWG. | External Wiring<br>All terminal blocks(as well as relay) shall be provided with 20% of wired extra terminal connectors<br>and 10% of extra terminal connectors, for future use.<br>Internal Wiring<br>All cable trays shall be sized in such a way that at maximum 60% of their width are<br>used, considering all spare and future I/O. | I-ET-3010.00-5520-888-P4X-001 | YES | | Noted | | Closed |
| 20 | **Gland plate** | 1 | Removable gland plate shall be provided by vendor.<br>1. Material<br>2) Hole arrangement<br>3. spare hole plugging<br>4. Earth jumper wire to equipment body. | Removable gland plate shall be provided by vendor.<br>1. Material: ASTM 316L stainless steel, with minimum thickness 2.5 mm.<br>2. Hole arrangement<br>3. Spare hole plugging<br>4. Earth jumper wire to equipment body. | I-ET-3010.00-5520-888-P4X-001 | YES | | Noted | | Closed |
| | | 2 | Removable Gland Plate shall not be painted and material shall be stainless steel, unless otherwise specified.<br>Gland plate size shall be enough to accommodate all connected cable gland based on HHI's cable information including spare holes. | If Removable Gland Plate is provided, it shall not be painted and material shall be stainless steel, unless otherwise specified.<br>Gland plate size shall be enough to accommodate all connected cable gland based on HHI's cable information including spare holes.<br>For outdoor panel,<br>Removable gland plates or Multi-cable Transits (MCT) shall be fitted for bottom cable entry. Gland plates shall be strong enough to support the necessary glands and cables. The panels shall have mounting legs of 300 mm height to accommodate installation and cable glands or MCTs. Panels shall be supplied with anti-vibration mats to be mounted between panel and mounting legs.<br>If more than two rows (see Figure 1) of cables are needed, cable glands shall not be used, only MCTs. | I-ET-3010.00-5520-888-P4X-001 | YES | | Noted | | Closed |
| 21 | **Wire** | 1 | Wire color, type(flame retardant..), size shall be designed as per project requirement. | Wire color, type(flame retardant..), size shall be designed as per project requirement.<br>External Wiring for instrument<br>• Made of non-hygroscope material;<br>• Adequate for conductors with 2.5 and 1.5 mm 2 cross section;<br>• Flame retardant;<br>• Insulated to 300 V;<br>• Of disconnect type (knife type).<br>Terminal block for elec. shall be suitable for yard interface cable.<br><br>Incoming and outgoing cables as well as internal wiring shall be supported and routed through plastic gray ducts with covers. In case ofintrinsically safe field wiring, the cables shall run through blue plastic ducts with protective cover.<br><br>Power supply cables shall run in cable trays segregated from the marshalling termination wiring, low voltage wiring and communication wiring. These terminals shall be covered with protective plastic covers and shall have warning signs. | I-ET-3010.00-5520-888-P4X-001 | YES | | Noted | ACCORDING TO OUR STANDARDS | Closed |
| 22 | **Bay requirement** | 1 | If internal wiring between panels to be dismantled due to delivery or installation at yard, it shall be specified on DWG and loosed wiring with wire markers/lugs shall be kept inside one of panels. | If internal wiring between panels to be dismantled due to delivery or installation at yard, it shall be specified on DWG and loosed wiring with wire markers/lugs shall be kept inside one of panels. | N/A (HHI Standard) | | NO | | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 23 | **Network** | 1 | All required components for Customer network interface shall be supplied by VENDOR.<br>1) FO Convertor<br>2) Patch code<br>3) Patch panel<br>4) Switch | 1 Gb/s shall be the default speed for all automation network, except for the SOS LAN ring, the CSS LAN ring, Package Units LAN ring, where the default speed shall be 10 Gb/s.<br>Switches shall possess 30% spare ports. (30% port shall applied to FO convertor as well)<br>All network devices shall be managed. (managed type switch)<br>All switches shall have port mirroring capabilities in all of its ports (this is required in order to allow the IDS solution to work). | I-MD-3010.1M-5520-800-P4X-003 | | NO | | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 24 | **In Skid Control panel & Remote Control panel** | 1 | The location of control panel shall be determined as per project requirement. | "VENDOR to bid based on all control panel shall be installed at Equipment skid edge excepting EDG remote control panel, inert gas gen remote control panel and marine growing protection system control panel.<br>In addition, all cabling between JB & elec devices in skid and control panel shall be completed under vendor scope.<br>If required during detailed engineering, control panel shall be supplied as ship loose and all cabling up to skid edge junction box(Vendor scope) shall be completed according to HHI request without cost variation" | I-MD-3010.1M-5520-800-P4X-003 | YES | | Noted | | Closed |
| 25 | **System requirement (Incl Training)** | 1 | Package Units shall also follow the requirements established for the whole Automation and Control system, with the exceptions foreseen in the aforementioned specification.<br>PACKAGER shall submit for PETROBRAS approval the detailed training program. In detailed program shall be presented the course plan, curriculum vitae of instructors, detailed agenda. | Package Units shall also follow the requirements established for the whole Automation and Control system, with the exceptions foreseen in the aforementioned specification.<br>PACKAGER shall submit for PETROBRAS approval the detailed training program. In detailed program shall be presented the course plan, curriculum vitae of instructors, detailed agenda. | I-MD-3010.1M-1200-940-P4X-025<br>I-MD-3010.1M-1200-800-P4X-001 | YES | | Noted | | Closed |
| **4. Test requirement** | | | | | | | | | | |
| 1 | **Internal test report** | 1 | Internal test report stamped by vendor QM/QC shall be provided with FAT notification.<br>All test report shall be inline to FAT procedure. | Internal test report stamped by vendor QM/QC shall be provided with FAT notification.<br>All test report shall be inline to FAT procedure. | N/A (HHI Standard) | YES | | Noted | | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) | | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Yes | No | | | |
| 2 | **IFAT** | 1 | IFAT shall be conducted with third party as per project requirement. | For IFAT between CSS and PACK. CSS Supplier will visit PKG vendor shop with test tools.<br>PKG VENDOR shall notice detailed test plan 4 weeks ago with readiness check sheet.<br>For PACKAGE FAT, CONTRACTOR shall provide technical assistance and a portable unit (notebook) fully configured with all the necessary software of SOS, properly configured, in order to test memory maps and HMIs functionalities between PACKAGE<br>and SOS.<br>* IFAT failure;<br>If major or excessive deviations from the requirements are detected during IFAT, HHI may choose to immediately halt the test and depart Supplier's facilities. If HHI halts the test and departs Supplier's facilities, Supplier shall be responsible for the following:<br>• Immediate completion of all internal testing, correction of deficiencies, and rescheduling of the test. Supplier shall be responsible for all costs associated with these corrective activities.<br>• Payment to HHI of purchase order and contract penalties should there be delays in shipments and delivery dates.<br>• 100% of CONTRACTOR/COMPANY's and CONTRACTOR/COMPANY representative's travel and living costs for attending the abandoned tests.<br>• 100% of all costs for staging and conducting the rescheduled IFAT.<br>• If partial delevery is required due to IFAT delay, it is carried out under supplier's responsibility as per CONTRACTOR's request. | I-MD-3010.1M-1200-800-P4X-001 | YES* | | Noted | FAT ACCORDING TO AGREED PROTOCOL.<br><br>ONLY CLASS SOCIATY EXPENSES ARE INCLUDED IN THE SCOPE.<br><br>CONTRACTOR`S REPRESENTATIVES ARE INVITED TO FAT, ALTHOUGH ALL TRAVELLIN G AND LIVING EXPENSES MUST BE COVERED BY THE CONTRACTOR | Closed |
| 3 | **SIL** | 1 | SIL verification / validation to be conducted as per project requirement. | SIL certification for safety devices shall be supplied | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 4 | **Heat soak test** | 1 | Heat soak test for electrical panel shall be carried out as per project requirement if applicable. | HOLD | HOLD | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 5 | **FAT** | 1 | FAT shall be conducted with third party as per project requirement | Inspection and test requirement in I-ET-3010.00-5140-700-P4X-003 clause 7 to be fulfiled.<br>PKG VENDOR shall notice detailed test plan 4 weeks ago with readiness check sheet.<br>* FAT failure;<br>If major or excessive deviations from the requirements are detected during FAT, HHI may choose to immediately halt the test and depart Supplier's facilities. If HHI halts the test and departs Supplier's facilities, Supplier shall be responsible for the following:<br>• Immediate completion of all internal testing, correction of deficiencies, and rescheduling of the test. Supplier shall be responsible for all costs associated with these corrective activities.<br>• Payment to HHI of purchase order and contract penalties should there be delays in shipments and delivery dates.<br>• 100% of CONTRACTOR/COMPANY's and CONTRACTOR/COMPANY representative's travel and living costs for attending the abandoned tests.<br>• 100% of all costs for staging and conducting the rescheduled FAT.<br>• If partial delevery is required due to FAT delay, it is carried out under supplier's responsibility as per CONTRACTOR's request.<br>All test specified on I-MD-3010.1M-1200-800-P4X-001 shall be conducted<br><br>Control, safety and monitoring equipment and devices (power supplies, workstations, microcomputers, data servers, communication cards, controllers, I/O racks and network devices, among others) shall be designed to operate satisfactorily under the environmental conditions stipulated by the Classification Society and project documents. | I-ET-3010.00-5140-700-P4X-003<br>I-MD-3010.1M-1200-800-P4X-001<br>I-ET-3010.1M-1200-800-P4X-001 | YES* | | Noted | FAT ACCORDING TO AGREED PROTOCOL.<br><br>ONLY CLASS SOCIATY EXPENSES ARE INCLUDED IN THE SCOPE.<br><br>CONTRACTOR`S REPRESENTATIVES ARE INVITED TO FAT, ALTHOUGH ALL TRAVELLIN G AND LIVING EXPENSES MUST BE COVERED BY THE CONTRACTOR | Closed |
| 6 | **Cyber security** | 1 | Cyber security verification / validation shall be conducted with third party as per project requirement. | CYBERSECURITY TESTING<br>Two Cybersecurity tests shall be foreseen in order to attest that the measures of this document (especially Sections 6 and 7_I-MD-3010.1M-5520-800-P4X-003) have been correctly implemented:<br>• Cybersecurity Factory Acceptance Test (CFAT)<br>o To be performed immediately after the Factory Acceptance Test, at the Automation System VENDOR facilities.<br>o Performed after functional testing of the system is complete<br>• Cybersecurity Site Acceptance Test (CSAT)<br>o Performed at the location of the system<br>o Performed after operational testing is complete<br>o Prior to releasing the system to operation<br>During these tests, a cybersecurity robustness testing shall also be performed. This test includes vulnerability analysis, penetration testing, and submitting the system to unusual network loads to ensure that it maintains its correct behavior.<br>*DETAILED ENGINEERING DESIGN DOCUMENTS specifed on I-MD-3010.1M-5520-800-P4X-003 _section 9 shall be supplied for review/approval.<br>NETWORK ONE-LINE DRAWING<br>DETAILED NETWORK INTERCONNECTION DIAGRAM<br>IP LIST<br>ASSET LIST<br>Other documents | I-MD-3010.1M-5520-800-P4X-003 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| **5. Numbering philosophy** | | | | | | | | | | |
| 1 | **Cable numbering** | 1 | Cable tag number shall be allocated as per project requirement. | Cable tag number shall be allocated as per project requirement. | I-ET-3000.00-1200-940-P4X-001 | YES | | Noted | | Closed |
| 2 | **Instrument numbering** | 1 | Instrument tag number shall be allocated as per project requirement. | Instrument tag number shall be allocated as per project requirement. | I-ET-3000.00-1200-940-P4X-001 | YES | | Noted | | Closed |
| 3 | **Hard IO numbering** | 1 | Hard IO number shall be allocated as per project requirement. | Hard IO number shall be allocated as per project requirement. | I-ET-3000.00-1200-940-P4X-001 | YES | | Noted | | Closed |
| 4 | **Soft IO numbering** | 1 | Soft IO number shall be allocated as per project requirement. | Soft IO number shall be allocated as per project requirement. | I-ET-3000.00-1200-940-P4X-001 | YES | | Noted | | Closed |
| 5 | **Loop numbering** | 1 | Instrument loop number shall be allocated as per project requirement. | Instrument loop number shall be allocated as per project requirement. | I-ET-3000.00-1200-940-P4X-001 | YES | | Noted | | Closed |
| 6 | **Tubing numbering** | 1 | Tubing tag number shall be allocated as per project requirement. | Tubing tag number shall be allocated as per project requirement. | I-ET-3000.00-1200-940-P4X-001 | YES | | Noted | | Closed |
| 7 | **Terminal block numbering** | 1 | Terminal block numbering within PKG shall be in line with project requirement. If it is not clear on project requirement, each vendor shall submit his proposal for approval before implementation. | Terminal block numbering within PKG shall be in line with project requirement. If it is not clear on project requirement, each vendor shall submit his proposal for approval before implementation. | HOLD | YES | | Noted | | Closed |
| 8 | **Wire marker numbering** | 1 | Wire marker numbering within PKG shall be in line with project requirement. If it is not clear on project requirement, each vendor shall submit his proposal for approval before implementation. | Wire marker numbering within PKG shall be in line with project requirement. If it is not clear on project requirement, each vendor shall submit his proposal for approval before implementation.<br>All panel wiring shall be identified in both ends, by plastic rings with the terminal block number. | HOLD | YES | | Noted | | Closed |
| 9 | **Junction box numbering** | 1 | Junction box numbering shall be allocated as per project requirement. | Junction box numbering shall be allocated as per project requirement. | I-ET-3000.00-1200-940-P4X-001 | | NO | | NOT APPLICABLE FOR OUR PACKAGE , NO JUNCTION BOXES | Closed |
| 10 | **Unit control panel (UCP) numbering / Local control panel(LCP) numbering** | 1 | UCP and LCP numbering shall be allocated as per project requirement. | UCP/LCP Numbering to be allocated as per project requirement | I-ET-3000.00-1200-940-P4X-001 | YES | | Noted | | Closed |
| 12 | **Elec. Equipment numbering** | 1 | All electrical equipment tag number shall be in line with project requirement. | All electrical equipment tag number shall be in line with project requirement. | I-ET-3000.00-1200-940-P4X-001 | YES | | Noted | | Closed |
| 13 | | 1 | | | | | | | | |
| **6. Others** | | | | | | | | | | |
| 1 | **Design Review meeting** | 1 | Each PKG E&I design review meeting shall be conducted as pe project requirement. | Each PKG E&I design review meeting for FAT shall be conducted 3 times as minimum.<br>Detail schedule to be confirmed during detail engineering. | N/A (HHI Standard) | YES | | Noted | | Closed |
| 2 | **SIL workshop** | 1 | PKG vendor shall attend SIL workshop as per project requirement if applicable. | Circular Letter CL-28, Q16<br>The only SIL/LOPA study that needs to be considered in EPC phase is regarding closed flare system. However, during Detail Design phase, if SELLER deems necessary to carry out a LOPA Analysis to any other system, BUYER shall be consulted.<br>Bidder to quote priced proposal for attendence of SIL workshop as an option. | N/A | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 3 | **Cyber security workshop** | 1 | PKG vendor shall attend cyber security workshop as per project requirement if applicable. | Bidder to quote priced proposal for attendence of cyber security workshop as an option. | N/A | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 4 | **Special cable scope** | 1 | 1. Special cable shall be supplied by PKG vendor. Maximum length from skid to control/electrical room is 200m for fixed platform and 350m for FPSO.<br>2) All cable type which is not specified in project cable specification shall be considered as special cable. | 1) Special cable shall be supplied by PKG vendor. Maximum length from skid to control/electrical room is 200m for fixed platform and 350m for FPSO. Length to be confirmed during detail engineering.<br>2) All cable type which is not specified in project cable specification shall be considered as special cable.<br>3) The cables of submerged motors must be suitable for installation in the seawater, and / or liquid hydrocarbons environment; cables must be designed to withstand this specific environment with degradation.<br>These cables must be supplied and installed on the motor (at factory) by the motor manufacturer, with the cable length specified in the pump / motor data sheet. | N/A (HHI Standard) | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |

# Technical Compliance Sheet

| | | | |
|---|---|---|---|
| PROJECT : | | P-78 | |
| RFQ No.: | | HQ-54 | |
| EQUIPMENT NAME : | | OILY WATER SEPARATOR | |
| INSPECTED ON : | | | |
| INSPECTED BY : | | | |

| Section No | Subjects | | Technical requirement | Detail Project Requirement ( Fill out by project team) | Relevant document number, If applicable ( Fill out by project team) | Comply (Vendor to fill out) | | HHI 1st Comment | Remark | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Yes | No | | | |
| 5 | **Standardization** | 1 | If required, the stardarization shall be applied to PKG accoding to project requirement. | I-ET-3010.00-1200-956-P4X-002 Clause 4.1.1<br>For field instruments and instrumentation accessories where painting is required I-ET-3010.00-1200-956-P4X-002 – GENERAL PAINTING shall be followed. Instruments of the same type and function shall be of the same manufacturer.<br><br>(HOLD)HHI considers that the Standardization Requirements on Field Instruments need not be applied to the inside of package when considering the cost impact. KSL to review and confirm.<br><br>I-ET-3010.00-5140-700-P4X-002 Clause 3.6.1<br>"Manufacturers shall keep uniformity of equipment and components for the same supply. The same model for lighting fixtures, sockets, junction box and all bulk material shall be used in all Unit modules."<br><br>(HOLD)Standardization is principally required for all bulk materials but the applicable standardization shall be clarified such as cables, etc.<br>However, HHI considers that the Standardization Requirements on Field Electrical devices (lighting fixtures, sockets, junction box and all bulk material which are located outside package) don't need to be applied to the inside of package when considering the cost impact caused by late selection of Vendor for aformentioned electrical components. KSL to review and confirm.<br><br>I-ET-3010.00-1200-800-P4X-002 Clause 6.4.4<br>P2x Supervisory software shall be based on Microsoft Windows® operational system and shall present information through graphic screens. Preferably, P2x Supervisory Software shall be of the same manufacturer and version as the UNIT SOS. PACKAGE SUPPLIER shall liaise with CONTRACTOR in order to identify UNIT SOS Supervisory Software manufacturer and version.<br><br>(HOLD) UNIT SOS Supervisory Software manufacturer and version shall be confirmed by KSL on an urgent basis in order to inform the related PACKAGERS of the same information before TBE / PO phase. | I-ET-3010.1M-1200-800-P4X-005<br>I-ET-3010.00-5140-700-P4X-002<br>I-ET-3010.00-1200-800-P4X-002 | | NO | Noted | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 6 | **Approved vendor for subvendor** | 1 | All subvendor shall be selected among project approved vendor list. If approved vendor is not available for certain items, it shall be submitted for approval. | All subvendor shall be selected among project approved vendor list. If approved vendor is not available for certain items, it shall be submitted for approval. | N/A (HHI Standard) | | NO | | WE HAVE SUBMITTED OUR SUB-SUPPLIER LISTS FOR YOUR REVISION. MOST OF THEM ARE INCLUDED PROJECT VENDOR LIST | Closed |
| 7 | **L/L review meeting** | 1 | 1. Lesson Learned meeting shall be conducted as per project requirement.<br>2) Compliance sheet for L/L shall be submiitted. | 1) Lesson Learned meeting shall be conducted during detail engineering.<br>2) Compliance sheet for L/L shall be submiitted during detail engineering | N/A (HHI Standard) | YES | | Noted | IF APPLICABLE | Closed |
| 8 | **Machine monitoring system interface** | 1 | Machine monitoring system shall be provided as per project requirement if applicable. | Comply to I-ET-3010.00-5500-854-P4X-001 and each PKG specification. | I-ET-3010.00-5500-854-P4X-001 | | NO | | NOT APPLICABLE FOR OUR PACKAGE | Closed |
| 9 | **Asset management interface** | 1 | Asset management (monitoring) system interface shall be provided as per project requirement.<br>Ex) Hart mux…<br>All software configulation at field devices shall be conducted by PKG vendor before the delivery.<br>Ex) Instrument tag number configulation at transmitter. | The following field instruments of critical equipment, including also Packages, shall be integrated in the AMS to the full extent:<br>-All transmitters and valve positioners connected to CSS-PCS or CSS-HCS, including the ones that are part of P0 type Packager;<br>-All transmitters connected to CSS-PSD or CSS-HSD, including the ones that are part of P0 type Packager;<br>-All Fire and Gas sensors connected to CSS-FGS or CSS-HFGS, including ones that are part of P0 type packager;<br>-All transmitters and valve positioners connected to Packages type P2 and P2C;<br>-All Fire and Gas detectors connected to Packages, exception to those connected to an addressable system. In the latter case, those may be connected if capable.<br>NOTE 1: All instruments connected to AMS system shall comply with HART FOUNDATION.<br>NOTE 2: For Package type P2 and P2C, PACKAGER shall use its UCP controller with analog modules with HART capability allowing AMS data access through Single Ethernet link. In case UCP's controller I/O module does not possess HART capability, it shall be submitted to PETROBRAS approval the installation of HART multiplexers with Ethernet converter. (Including FO convertor, patch panel, patch cord)<br>A self-paced user training package shall be included with the AMS, plus details of the Manufacturer's scheduled training courses updated through website. | I-ET-3010.00-1200-850-P4X-002 | | NO | | NOT APPLICABLE FOR OUR PACAKGE | Closed |
| 10 | **Sunshade** | 1 | Sunshade shall be supplied as per project requirement. | All electrical/electronical components subjected to direct sun incidence shall have a cover, in order to minimize damage caused to excessive heating of the component's enclosure. | I-ET-3010.1M-1200-800-P4X-005 | | NO | | NOT APPLICABLE, PACKAGE LOCATED INDOORS | Closed |
| 11 | **Spare part** | 1 | Spare part shall be provided as per project requirement. | Packager shall provide a list of spare parts for all electrical equipment, for at least 2 (two) years of continuous operation, including prices and part number codes.<br>Packager shall provide all unusual tools necessary for maintenance, assembly or disassembly of all electrical equipment.<br>Packager shall provide the necessary spare parts for the commissioning and pre-operation periods. | I-ET-3010.00-5140-700-P4X-003 | YES | | Noted | | Closed |
| 12 | **FAT** | 1 | Equipment dimension and internal arrangement shall be inspected based on approved vendor GAD and signed off by Vendor QA prior to FAT. Discrepancy between equipment and vendor GAD shall be marked up in signed drawing. | Equipment dimension and internal arrangement shall be inspected based on approved vendor GAD and signed off by Vendor QA prior to FAT. Discrepancy between equipment and vendor GAD shall be marked up in signed drawing. | N/A (HHI Standard) | YES | | Noted | | Closed |

**Attachment** **1** E&I tie in document guidance